Downloaded from gsabulletin.gsapubs.org on October 28, 2010

Geological Society of America Bulletin

Paleomagnetic constraints on the tectonic evolution of Bare Mountain, Nevada

John A. Stamatakos, David A. Ferrill and Kathy H. Spivey

Geological Society of America Bulletin 1998;110;1530-1546
doi: 10.1130/0016-7606(1998)110<1530:PCOTTE>2.3.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Bulletin

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

Paleomagnetic constraints on the tectonic evolution of Bare Mountain, Nevada

John A. Stamatakos*
David A. Ferrill
Kathy H. Spivey
Center for Nuclear Waste Regulatory Analyses, Southwest Research Institute, San Antonio, Texas 78238

ABSTRACT

Paleomagnetic investigations of Tertiary dikes and Ordovician carbonate rocks at Bare Mountain and in the surrounding region of southwestern Nevada were conducted to further constrain possible rotations related to Walker Lane dextral shear and Basin and Range detachment faulting. Demagnetization experiments revealed two distinct remanence components in the Ordovician carbonate rocks: (1) a dual polarity Permian-Triassic magnetization carried by magnetite that was acquired after Paleozoic or Mesozoic folding but prior to Tertiary tilting; and (2) a dual polarity Miocene magnetization carried by hematite that was acquired after folding and tilting. The Tertiary dikes also contain a dual polarity magnetization carried by magnetite that can be interpreted either as a primary thermoremanent magnetization or a secondary chemical remanent magnetization acquired shortly after dike intrusion. Overall, mean directions for all three magnetizations roughly coincide with corresponding reference directions for Nevada. The largest declination discordance of 16° ± 7° (clockwise) comes from comparison of the Permian-Triassic magnetite magnetization to the Late Permian reference direction for Nevada. Thus, although structural and geothermal data, including zircon fission-track ages, indicate that northern Bare Mountain was tilted 20°–40° northeastward about a horizontal axis during early Tertiary time, we find no evidence for large-scale (i.e., >30°) vertical-axis rotation. On the basis of these results, we suggest that Bare Mountain forms an autochthonous or parautochthonous footwall within the Basin and Range and Walker Lane, in a region where many of the surrounding upper crustal rocks are detached and significantly rotated about vertical and steeply inclined axes. In addition, the nature and timing of footwall uplift indicated by our paleomagnetic results do not support tectonic models that reconstruct Bare Mountain and the Striped Hills as adjacent segments of a continuous fold and thrust belt system. Finally, we suggest that the postfolding Permian–Triassic magnetization at Bare Mountain coupled with Paleozoic zircon fission-track ages indicate that the Meiklejohn Peak thrust predates Jurassic-Cretaceous (Sevier) deformation and may have instead resulted from a Permian or earlier contractional event.

INTRODUCTION

Bare Mountain is an exhumed crustal inlier of folded and faulted Proterozoic and Paleozoic rocks situated in the southern Great Basin (Fig. 1A, p. 1535) within the central Basin and Range extensional province of Wernicke (1992). Bare Mountain is part of the diffuse belt of deformation known as the Walker Lane that consists of spatially segregated regions of east-west extension between northwest-trending zones of dextral strike-slip or oblique-slip faulting (Stewart, 1988; Oldow et al., 1994). Deformation within the Walker Lane and central Basin and Range is manifested by kilometer-scale translation of upper crustal blocks coupled with rotations about horizontal, steeply inclined, and vertical axes.

Vertical-axis rotations of crustal blocks within the Walker Lane have been documented and quantified by paleomagnetic results. For example, paleomagnetic directions identified as Cenozoic and Permian-Triassic secondary magnetizations in Paleozoic carbonate rocks and primary magnetizations in Cenozoic (mainly Miocene) clastic rocks and nonmarine limestones show as much as 100° of clockwise rotation associated with motion along the Las Vegas Valley shear (Gillett and Van Alstine, 1982; Nelson and Jones, 1987; Sonder et al., 1994). Significant clockwise rotations are also observed in the Black Mountains, presumably in response to similar right-lateral shear along the Death Valley–Furnace Creek fault zone (Holm et al., 1993), and in the Cottonwood Mountains, presumably in response to west-northwest Basin and Range extension (Snow et al., 1993).

Despite large rotations in many parts of the Walker Lane, paleomagnetic evidence for significant vertical-axis rotations in the region around Yucca Mountain and at Bare Mountain is equivocal. Primary magnetization directions from most of the middle Miocene volcanic rocks of the southwestern Nevada volcanic field are not systematically rotated (Hudson, 1992; Hudson et al., 1994), except within the southern reaches of Yucca Mountain (Rosenbaum et al., 1991) and in southwestern Crater Flat, just south of Bare Mountain (Hudson et al., 1996). Yet, preliminary paleomagnetic results from two Miocene dikes at Bare Mountain do not indicate any vertical-axis rotations of the Bare Mountain block (Zhang et al., 1993).

Structural and sedimentological evidence for large-magnitude vertical-axis rotations of the Bare Mountain block is also equivocal. Snow (1992; 1994), for example, envisioned the Panama and Meiklejohn Peak thrusts exposed at Bare Mountain as rotated segments of an Early Permian, pre-Sonoma thrust system that is also exposed in the Cottonwood Range in northern Death Valley. Reconstruction of the thrust system predicts large-scale translation along a Cenozoic regional detachment and a 90° clockwise vertical-axis rotation of Bare Mountain. Snow and Prave (1994) argued that paleoflow orientations of trough cross-bedding axes in the clastic Precambrian and Cambrian quartzites at Bare Mountain are rotated 85° ± 12° clockwise relative to regional paleoflow trends, consistent with their interpretation of ~90° clockwise rotation. In contrast, Caskey and Schweickert (1992) and Schweickert and Lahren (1997) viewed the Panama and Meiklejohn Peak thrusts as hindward

*E-mail: jstam@swri.edu.

GSA Bulletin; December 1998; v. 110; no. 12; p. 1530–1546; 11 figures; 4 tables.

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

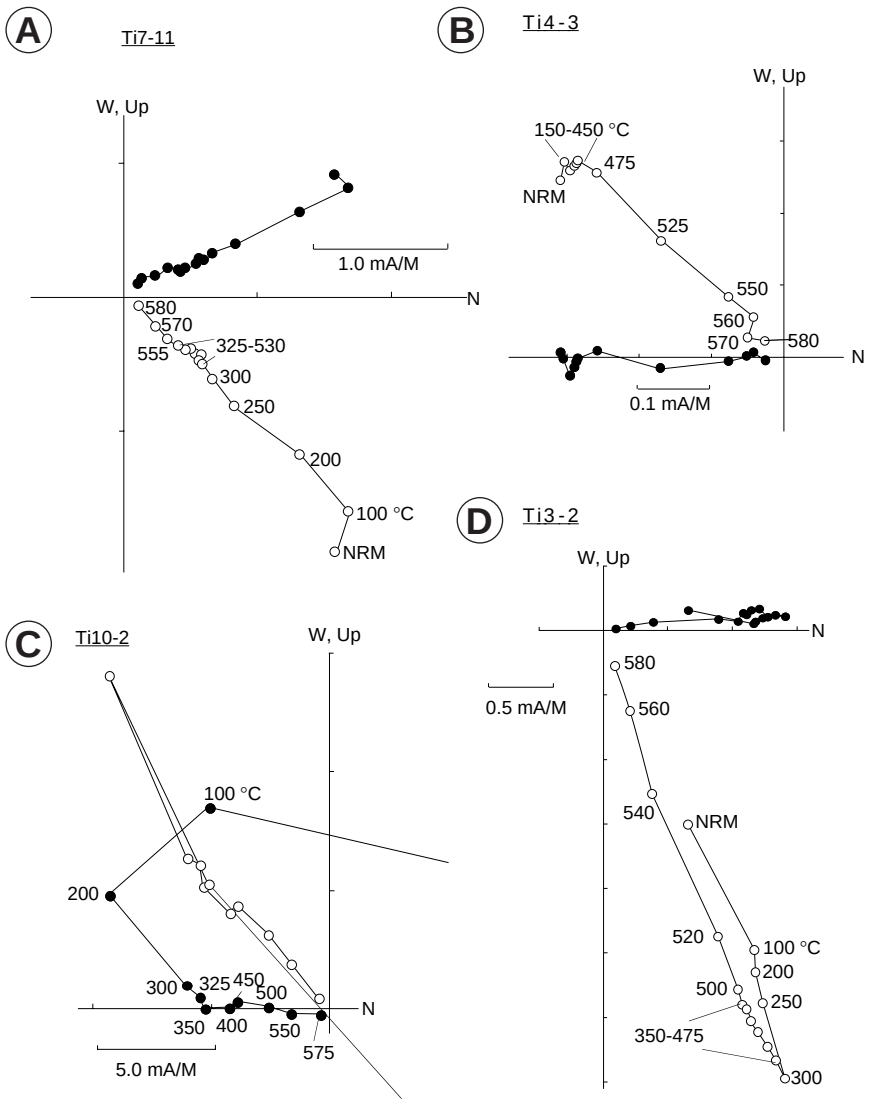


Figure 2. Sample vector end-point demagnetization diagrams (Zijderveld, 1967) showing thermal demagnetization of silicic porphyry dikes (A, B, and D) and diorite dikes (C) in in situ coordinates. Open and closed symbols are projections onto the vertical and horizontal planes, respectively. NRM—natural remanent magnetization.

and forward-verging features of the Mesozoic Sevier hinterland. In this model, the Amargosa Desert fault system, which includes the Bare Mountain fault, has presumably accumulated more than 30 km of dextral displacement since the mid-Miocene with minimal vertical-axis rotation of the Bare Mountain block.

This paper evaluates the possibility of large-scale vertical axis rotation of Bare Mountain related to Basin and Range extension and the Walker Lane transtensional deformation using a detailed and systematic study of the paleomagnetism of Bare Mountain, Nevada. Specifically, paleomagnetic results from 12 Oligocene(?) and Miocene mafic and silicic porphyry dikes and 20 sites collected in the carbonate rocks of the Ordovician Antelope Valley Formation at Bare Mountain are presented. These results are also examined within the context of the Walker Lane and Basin and Range tectonics by comparing them to paleomagnetic directions from 13 additional sites collected in the lower Paleozoic carbonate rocks exposed in nearby ranges of the Yucca Mountain region, as well as previously published paleomagnetic results from other ranges in Nevada and Utah.

GEOLOGIC SETTING AND SAMPLING

Stratigraphy

Bare Mountain exposes nearly 7.5 km of Proterozoic and Paleozoic stratigraphy of the Cordilleran miogeocline. The sequence represents the proximal (cratonward) facies of the margin consisting of late Precambrian to Cambrian clastic rocks (mainly quartzites and argillites) and Middle Cambrian to Devonian shallow-water platform carbonates overlain by Upper Devonian to Mississippian Antler foredeep deposits (e.g., Trexler et al., 1996). The stratigraphic record between Mississippian sedimentation and Miocene volcanism is not preserved at Bare Mountain; however, the mountain contains small-volume igneous intrusions that yield Cretaceous and Oligocene or Miocene radiometric ages (U-Pb data from zircon of 98 ± 27 Ma and K-Ar dates from hornblende of 26.1 ± 1.7 Ma and biotite of 16.6 ± 1.2 Ma as reported in Monsen et al., 1992). If Pennsylvanian and Permian rocks noted elsewhere in the Basin and Range and Mojave Desert were also at Bare Mountain, they have been eroded.

In southern Nevada, Miocene silicic volcanism associated with the southwestern Nevada volcanic field began about 16 Ma and continued until about 10 Ma (e.g., Sawyer et al., 1994). Near Bare Mountain, a nearly 2-km-thick sequence of pyroclastic deposits and rhyolitic lavas was deposited from a complex network of calderas to the north and northeast. Minor late Miocene to Holocene basaltic flows and dikes are also present, including Pliocene and Quaternary basaltic volcanic rocks of Crater Flat and the Amargosa Desert. The volcanic sequence is capped by latest Miocene, Pliocene, and Quaternary surficial deposits (Monsen et al., 1992). In the actively subsiding parts of the basins, the Pliocene-Quaternary volcanic rocks are partially or wholly buried by thick accumulations of Quaternary and Holocene alluvium and colluvium (Stamatakos et al., 1997). Along the eastern flank of Bare Mountain, a series of vertical, north-trending, 1–10-m-wide silicic porphyry dikes are exposed (Cornwall and Kleinhample, 1961; Carr, 1988; Noble et al., 1989). Conventional K/Ar and $^{40}Ar/^{39}Ar$ dates from dikes in the eastern Bare Mountain range in age from 14.9 to 13.8 Ma (Marvin et al., 1989; Noble et al., 1991; Monsen et al., 1992; Weiss, 1996).

Structure

In its present structural configuration, Bare Mountain constitutes the footwall of at least two Neogene extensional-fault systems, the Fluorspar Canyon detachment and the Bare Mountain fault. The Fluorspar Canyon detachment, as part of the

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

larger Bullfrog Hills detachment system, presumably unroofed Miocene volcanic rocks from northern Bare Mountain to the present location north and northwest of the mountain (Maldonado, 1990; Hoisch et al., 1997). The Bare Mountain fault is a north-south–trending, east-dipping normal fault with more than 2.5 km of offset that juxtaposes sedimentary and metasedimentary rocks against Pleistocene to Holocene deposits of the adjoining Crater Flat basin (e.g., Ferrill et al., 1996a; Fridrich, 1998). Because of the possibility for large-magnitude earthquakes, the Bare Mountain fault is one of the structures that is critical to safety evaluations and overall performance assessments of the proposed high-level nuclear waste repository at nearby Yucca Mountain.

In addition to these two Neogene extensional fault systems, Bare Mountain reveals geologic evidence of one or more Tertiary (pre-Miocene) extension events. Structural and geothermometric results (Stamatakos and Ferrill, 1996) suggest that Bare Mountain was exhumed and rotated about a west-northwest horizontal axis between Eocene and middle Miocene time, prior to extension along the Fluorspar Canyon detachment. The cause of this horizontal-axis rotation and exhumation is unknown. This tilting event, however, was itself preceded by down-to-the-southeast extension along a series of normal faults that dissect the northwestern corner of the Bare Mountain block, including the ~2.5 km of displacement currently observed across the Gold Ace Mine fault (Ferrill et al., 1998).

Bare Mountain also preserves evidence of Mesozoic or possibly late Paleozoic contractional deformation. In the northeast corner of Bare Mountain, the Meiklejohn Peak thrust places Lower Cambrian through Lower Devonian carbonate rocks and quartzites on Devonian-Mississippian Antler foredeep strata along a duplex of southeast-verging (in present coordinates) thrust faults (Fig. 1B, p. 1536). In southern Bare Mountain, the Panama thrust places Precambrian quartzites northwestward (in present coordinates) on top of a large footwall syncline of latest Precambrian and Cambrian quartzites.

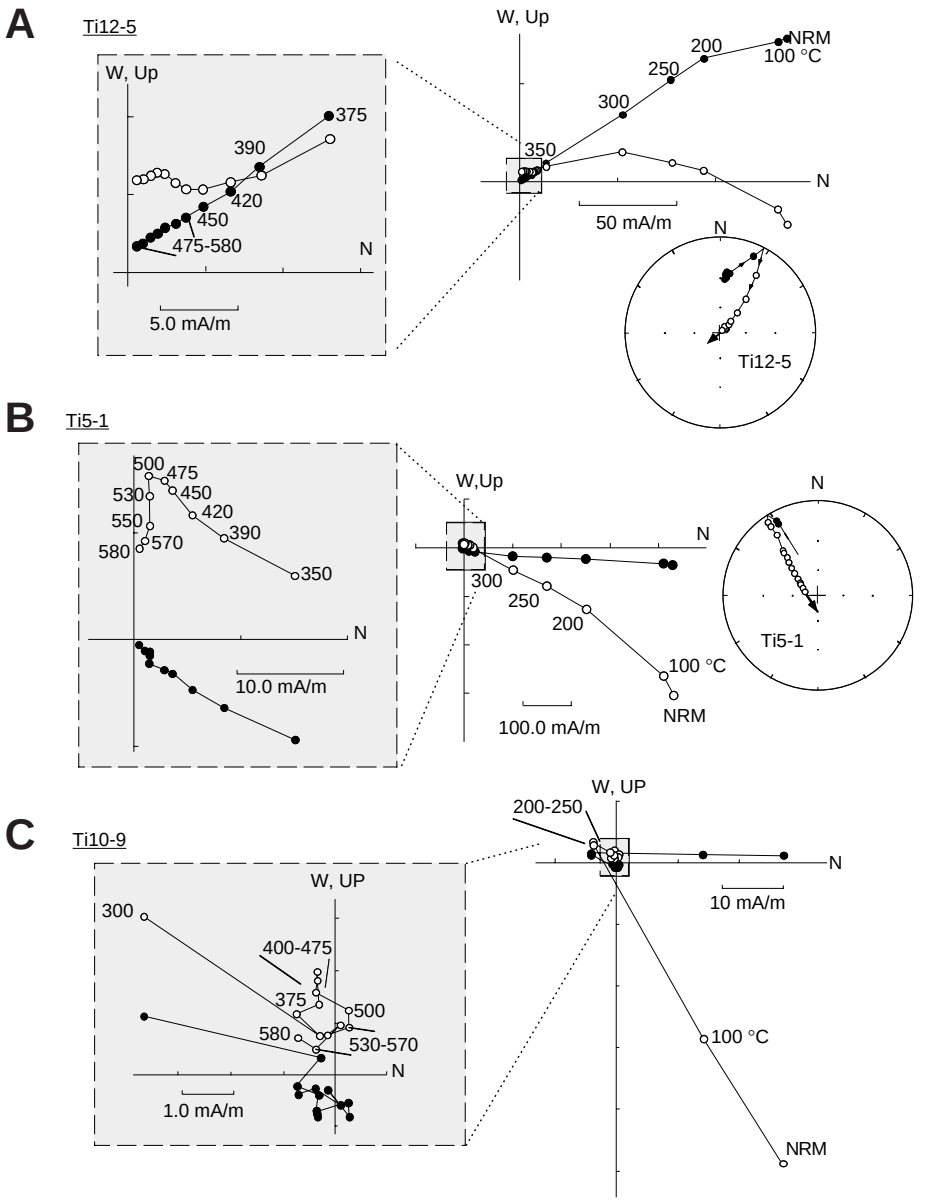


Figure 3. Sample vector end-point demagnetization diagrams and corresponding equal-angle projections showing remagnetization circle behavior for dike samples characterized by normal and reversed polarity components with strongly overlapping coercivity and unblocking spectra. In the vector end-point diagrams, the open and closed symbols are the projection onto the vertical and horizontal planes, respectively. In the equal-angle projections, open and closed symbols are projections onto the upper and lower hemispheres, respectively. NRM—natural remanent magnetization.

Thermal History

Metamorphic grade and peak geothermal (burial) temperatures decrease from west and southwest to the northeast across Bare Mountain. Amphibolite facies rocks are exposed west of the Gold Ace Mine and north of the Conejo Canyon faults, and subgreenschist and nonmetamorphic rocks are exposed elsewhere (Monsen et al., 1992). Peak burial temperatures estimated from calcite-twin geothermometry (Stamatakos and Ferrill, 1996), conodont alteration indices (Grow et al., 1994), and isotopic and fission-track thermochronologies (Hoisch et al., 1997; Ferrill et al., 1996b) indicate variations from greater than 350 °C in the northwest corner of Bare Mountain to less than 150 °C in the hanging wall of the Meiklejohn Peak thrust (Fig. 1B, p. 1536). Geothermometric constraints as well as fold plunges and fault-bedding intersections show that Bare Mountain developed this structural plunge prior to apatite fission-track ages being set in middle Miocene time (Ferrill et al., 1996b). In addition, textural features and K-Ar dates from alunite indicate significant epithermal and hydrothermal activity penecontemporaneous with the later stages of silicic volcanism (ca. 11.0–12.5 Ma), especially in association with altered margins of the silicic porphyry dikes (Weiss, 1996; Noble et al., 1991).

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

TABLE 1. PALEOMAGNETIC RESULTS FROM MIOCENE SILICIC PORPHYRY DIKES AT BARE MOUNTAIN, NEVADA

			In situ		Preplunge			
Site	n/N	R	Dec	Inc	Dec	Inc	κ	α_{95}
Til	0/16	0						
Ti2	12/12	7	161.8	–60.1	180.0	–34.5	11.3	13.5
Ti3	6/6	0	355.8	66.3	18.7	37.4	39.7	10.8
Ti4	7/11	7	184.4	–46.6	194.8	–17.5	149.7	4.9
Ti4 bc	0/9	0						
Ti5	0/11	0						
Ti6	0/12	0						
Ti7	9/13	6	156.5	–39.6	172.4	–19.9	136.8	4.4
Ti8	9/14	0	343.5	44.5	359.8	21.7	6.9	21.2
Ti9	12/12	1	154.4	–44.7	174.1	–24.9	21.3	9.6
TilO	10/13	8	176.2	–60.3	195.6	–32.0	23.4	10.9
Til 1	0/9	0						
Til2	2/11	2	(180.0)	(–39.0)	(194.3)	(–9.7)		
Mean	7/12	5	**166.5**	**–52.2**	185.8	–27.4	**42.4**	**9.4**

Notes: n is the number of samples that produced CHRM directions, and N is the number of samples from the site demagnetized. R is the number of samples with reversed CHRM directions. Dec and Inc are the declination and inclination of the site mean direction, in degrees. κ is the Fisher (1953) precision parameter, and α95 is the 95% confidence region about the mean value. Ti4bc is the baked contact test at site Ti4. Values in parentheses were not used in overall mean calculations. Bold values indicate best estimate of grand mean direction for the Tertiary dikes.

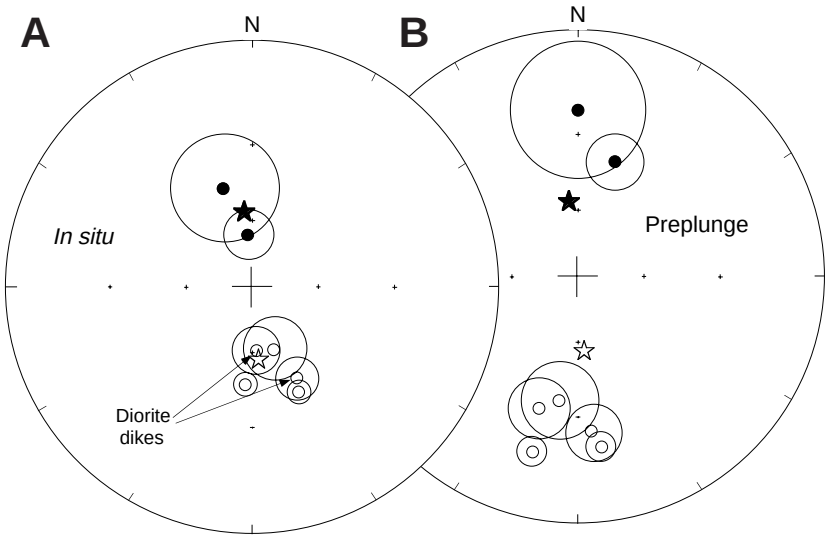


Figure 4. Equal-angle projections showing site mean directions and associated 95% cones of confidence for Tertiary silicic porphyry and diorite dikes before (A) and after (B) correction for structural plunge at Bare Mountain. Solid and open stars show the positions of the normal and reversed polarity reference directions, respectively, for Nevada, based on the Miocene reference pole for North America (Hagstrum et al., 1987). Open and closed symbols are projections onto upper and lower hemispheres, respectively.

Sample Collection

At Bare Mountain, samples for paleomagnetic analyses were collected from 12 sites in the Tertiary silicic porphyry and diorite dikes. These correspond to six dikes mapped by Monsen et al. (1992) and six dikes not shown on their map (sites Ti4, Ti6, Ti8, Ti9, Ti11, and Ti12) (Fig. 1B, p. 1536). At site Ti4, samples were also collected and analyzed from the altered country rock adjacent to the dike walls with the intent of performing a baked contact test. We sampled 20 sites in the Ordovician Antelope Valley Formation, seven sites from the hanging-wall anticline of the Meiklejohn Peak thrust, and 13 sites within the footwall rocks of the Meiklejohn Peak thrust (Fig. 1B, p. 1536). We also collected 13 sites of Ordovician Pogonip Group carbonate rocks from other localities within the Yucca Mountain region (Fig. 1A, p. 1535), including 5 sites from the Striped Hills, 2 sites each from the Specter Range, Yucca Flat, and Pahrump Valley, and 1 site each from immediately north of Mercury, Nevada, and in the Black Hills between the Nevada Range and the Las Vegas Range. Paleomagnetic cores were either drilled in the field with a gasoline-powered and water-cooled coring device or collected as oriented hand samples and cored in the lab. Sample orientations and bedding attitudes were measured using a magnetic compass after field checks verified that magnetizations of the rocks did not bias compass readings. Between 6 and 16 samples were collected at each site.

METHODS

Characteristic and secondary magnetizations were isolated from progressive thermal demagnetization of the samples. Standard thermal demagnetization was supplemented with alternating field (AF) followed by thermal demagnetization, and acquisition of anhysteretic remanence and isothermal remanence magnetizations to provide additional information on magnetic mineralogy. Thermal demagnetizations were conducted using a standard paleomagnetic furnace with a peak temperature of 700 °C and a maximum rest field of 10 nT. An AF demagnetizer with a peak field of 200 mT coupled with an automated internal coil capable of generating 1G biasing fields were used for AF demagnetization and anhysteretic remanence acquisition experiments. An electromagnet capable of generating static magnetic fields to 1.3 T was used for isothermal remanence magnetization acquisition experiments. All magnetic measurements were made using a three-axis cryogenic magnetometer in the magnetically shielded facility at the University of Michigan. Peak ambient field in the shielded room is less than 600 nT.

Magnetic components isolated during demagnetization experiments were determined from visual inspection of the linear segments on vector end-point diagrams. Component directions were determined from a least square fit of the observed linear trajectories (e.g., Kirschvink, 1980). Sample directions were combined to form site means using simple vector addition, giving unit weight to each sample direction. Statistical parameters were calculated from the method of Fisher (1953). Where possible, fold tests (Graham, 1949) were

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

used to bracket the age of the magnetizations. Fold-test statistics were based on the McFadden and Jones (1981) and McElhinny (1964) methods. In areas where the magnetic declinations suggest significant vertical-axis rotations, an inclination-only fold test was used (Parés et al., 1994), following the method of McFadden and Reid (1982). Calculation of the statistical significance of tectonic rotations based on paleomagnetic directions was from Demarest (1983).

Two structural corrections were made during analysis of the paleomagnetic directions, one for the tilting of the entire Bare Mountain and Striped Hills blocks related to Tertiary (but pre–middle Miocene) detachment faulting, and one for folding of the Paleozoic strata from Mesozoic or late Paleozoic Cordilleran contractional deformation. For the plunge correction, site mean directions were tilted about a horizontal axes. The plunge axis of the Meiklejohn Peak anticline (trend = 039°, plunge = 33° northeast) was used as the reference for this correction at Bare Mountain. A plunge determined from bedding data (trend = 087°, plunge = 30° east) was used as a reference line for plunge corrections of site means at the Striped Hills. In the fold tests, magnetization components from folded Paleozoic carbonates were rotated about their respective lines of strike in 5% increments of bedding dip after both magnetic directions and bedding orientations were corrected for respective structural plunges.

A third structural correction, that for footwall tilting of the Bare Mountain block from Miocene to recent motion along the Bare Mountain fault, was too small to be of consequence. Geometric considerations (i.e., fault dip, width of the footwall block, and cumulative fault displacement) indicate a maximum correction of 7° west about a roughly north-south axis. Map patterns (Monsen et al., 1992), however, show that the silicic porphyry dikes are essentially vertical, suggesting that, if the dikes were intruded into Bare Mountain vertically, the actual tilting of Bare Mountain since the Miocene was less than 5°, and possibly as small as 2°—an observation consistent with the 1°–3° difference in slope angle of alluvial fans on the eastern and western flanks of Bare Mountain (Ferrill et al., 1996a, 1997).

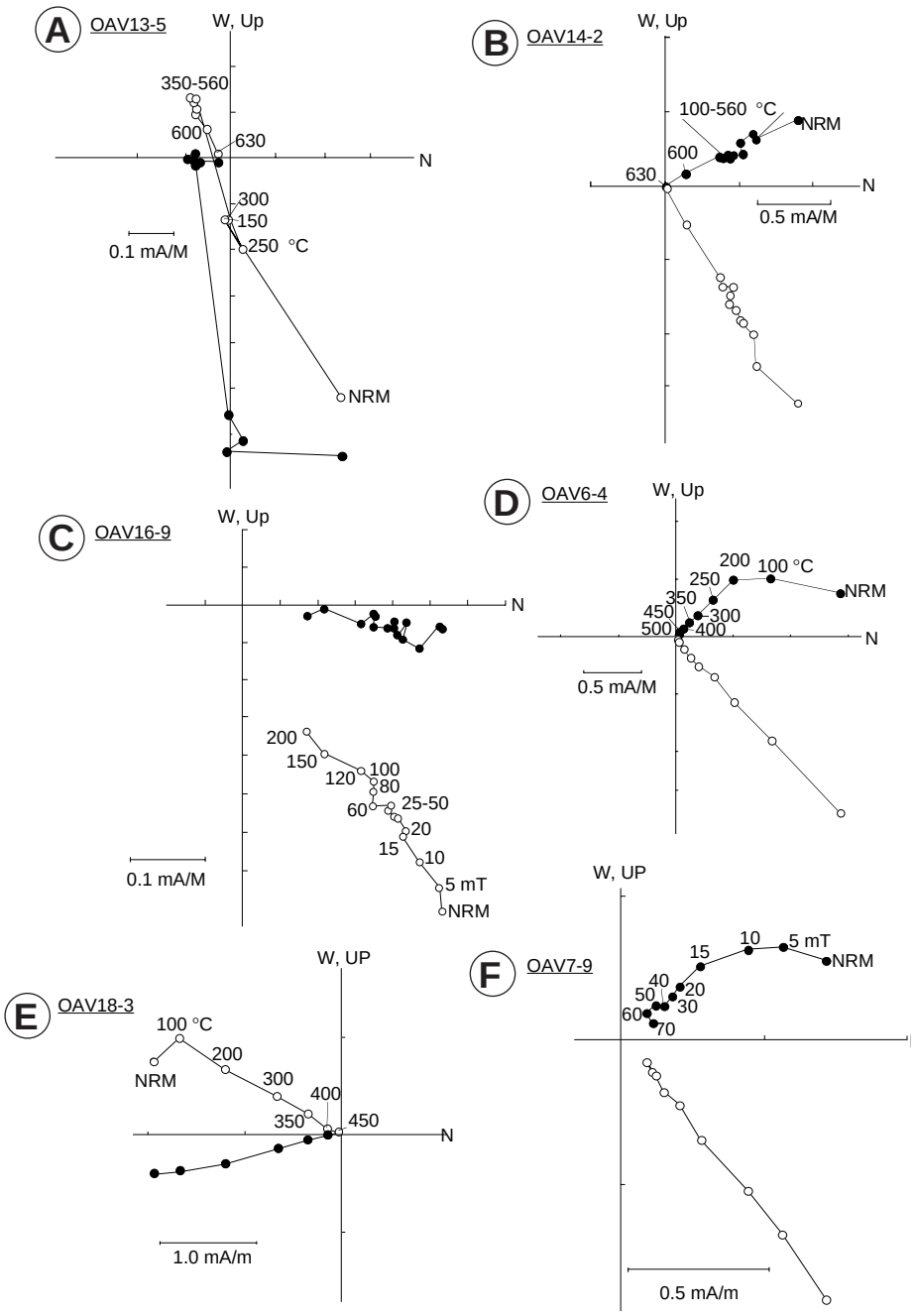


Figure 5. Sample vector end-point demagnetization diagrams showing thermal and alternating field demagnetization (in mT) of the Ordovician Antelope Valley Formation carbonate rocks at Bare Mountain, Nevada. Samples from the footwall of the Meiklejohn Peak thrust (A, B, and C) have high coercivity and relatively high unblocking temperature spectra. Samples from the hanging wall of the Meiklejohn Peak thrust (D, E, and F) have moderate coercivity and unblocking spectra. Open and closed symbols are projections onto the vertical and horizontal planes, respectively. NRM—natural remanent magnetization.

PALEOMAGNETIC RESULTS

Silicic Porphyry and Diorite Dikes

Intensities of the natural remanent magnetizations range between 0.1 mA/m and 15 A/m, although most samples (~95%) yielded natural remanent magnetization intensities between 1.0 and 100.0 mA/m. The remaining 5% of the samples with either extremely strong (probably from lightening strikes) or weak (probably from extensive hydrothermal or epithermal alterations) natural remanent magnetization intensities did not produce coherent and stable demagnetizations. The demagnetization trajectories are erratic, with unsystematic changes in sample intensity and direction. Samples from the altered country rock adjacent to the dike wall at site Ti4 also exhibited unstable demagnetization behavior, in which nearly all of an apparent randomly oriented natural remanent

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

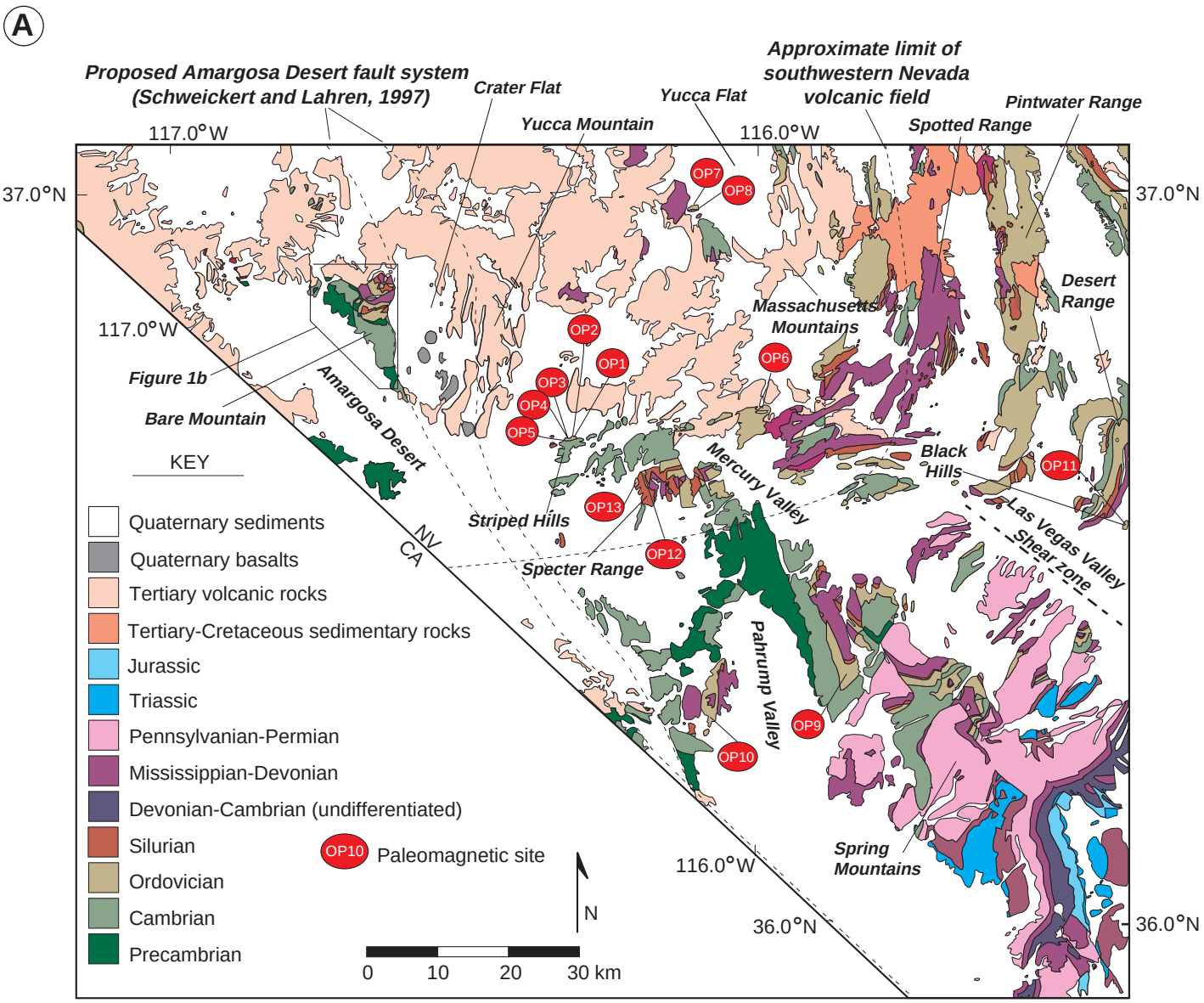


Figure 1. (A) Geologic map of southwestern Nevada based on Longwell et al. (1965), Tschanz and Pampeyan (1970), Albers and Stewart (1972), and Cornwall (1972), showing the location of Bare Mountain and distribution of paleomagnetic sites in Ordovician Pogonip Group carbonate rocks.

magnetization signal was removed within the first 200 °C of heating.

Pilot thermal and AF experiments indicated that thermal demagnetization was the more effective demagnetization technique for these samples. In the pilot demagnetization experiments, two distinct types of demagnetization behavior were observed: (1) those with a single magnetization component or two distinct magnetization components consisting of discrete unblocking or coercivity spectra; and (2) those with strongly overlapping unblocking and coercivity spectra consisting of at least two magnetization components.

About half the samples showed stable univectorial decay of the natural remanent magnetizations (Fig. 2, A and B) or discrete unblocking of two antipodal components (Fig. 2, C and D). Both north and moderately down or south and moderately up characteristic remanent magnetization directions were obtained. In samples with both polarity components, examples were found in which the north and down component overprints a south and up direction (Fig. 2C), and opposite cases in which the south and up component overprints a north and down component (Fig. 2D). Maximum unblocking temperatures for all samples were consistently near 580 °C. The median destructive field for sample AF demagnetizations averaged about 20 mT.

In the other half of the samples, the unblocking temperature and coercivity spectra of the two antipodal components overlap such that single component characteristic remanent magnetization directions could not be isolated. In many of the samples with this type of behavior, overlapping coercivity and unblocking temperature spectra persisted until the signal was essentially demagnetized (Fig. 3, A and B). More erratic demagnetization behavior was occasion-

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

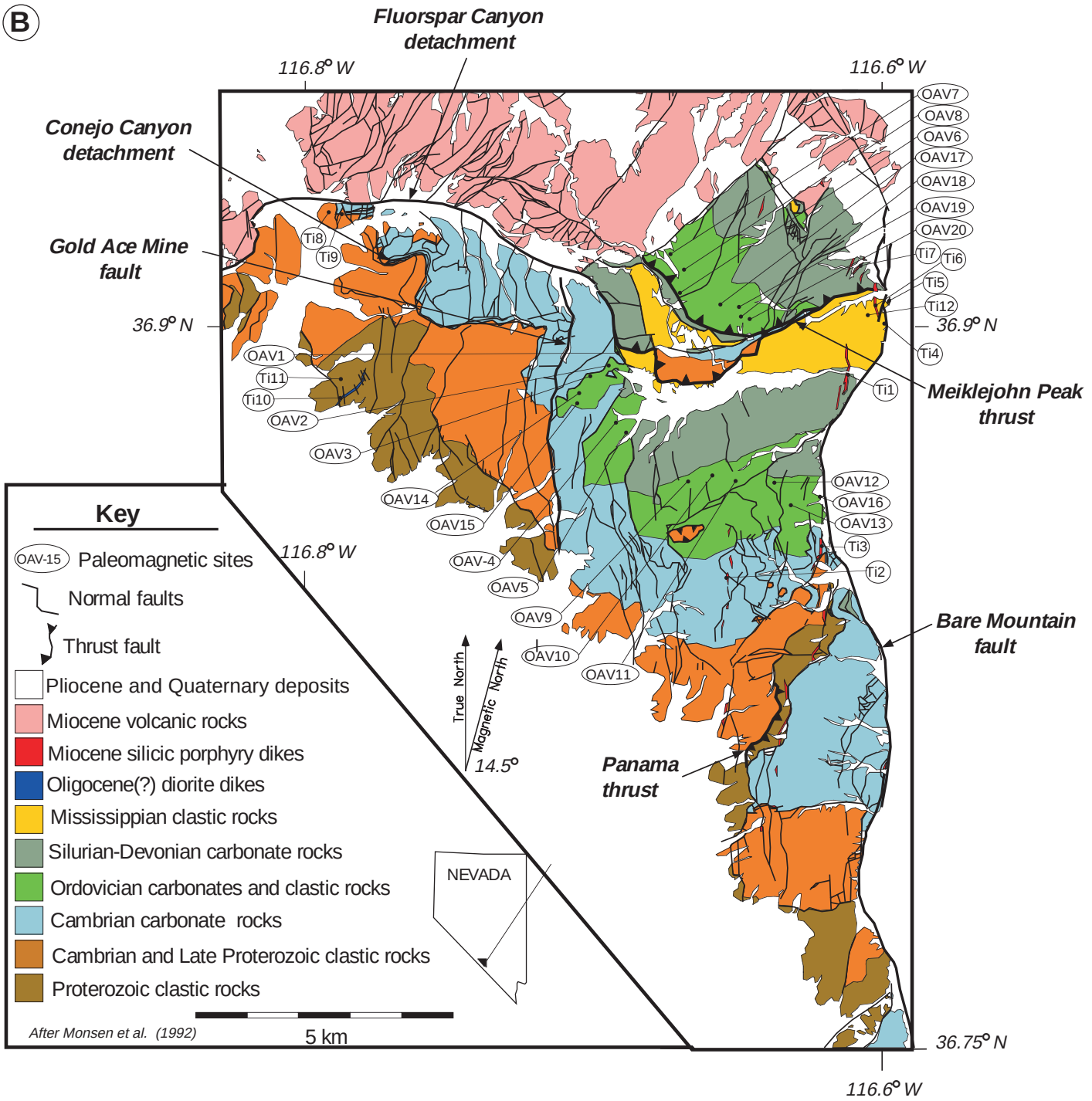


Figure 1. (*Continued*). (B) Geologic map of Bare Mountain, Nevada, simplified from Monsen et al. (1992), showing distribution of paleomagnetic sites in the Ordovician Antelope Valley Formation carbonate rocks and Tertiary silicic and diorite dikes.

ally observed after removal of a strong north and down overprint (Fig. 3C). Remagnetization circles follow great-circle paths between antipodal north and down and south and up characteristic remanent magnetization directions isolated from samples with more discrete single or two component magnetizations. Remagnetization circle analyses were not attempted in these samples because sufficient results were obtained from samples with stable, discrete-component magnetizations.

Site mean directions were obtained for 7 of the 12 sites, 5 from the silicic porphyry dikes and 2 from the diorite dikes (Table 1). The remaining sites were dominated by the remagnetization circle behavior described previously, hence no site mean directions were computed. Two stable sample directions were obtained for site Ti12 consis-

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

TABLE 2. PALEOMAGNETIC RESULTS FROM THE ORDOVICIAN ANTELOPE VALLEY FORMATION, BARE MOUNTAIN, NEVADA

Site	DD/D	DD′/D′	n/N	R	In situ Dec	In situ Inc	Preplunge Dec	Preplunge Inc	Stratigraphic Dec	Stratigraphic Inc	κ	α_{95}
Hematite Magnetization (Footwall)												
OAV1	357/14	242/24	5/8	1	4.1	62.0	20.2	32.1	2.7	51.8	44.4	11.6
OAV2	006/22	262/22	7/8	0	6.8	52.3	18.4	22.4	8.1	30.2	33.5	10.6
OAV3	019/14	233/20	3/8	0	(–36.6)	(42.7)	(37.2)	(9.7)	(35.2)	(28.9)	(16.2)	(31.4)
OAV4	048/34	121/05	3/8	0	1.5	57.1	17.0	28.1	19.7	29.6	128.8	10.9
OAV5	335/30	276/32	2/8	0	–0.4	(37.5)	(8.5)	(10.1)	(2.7)	(10.0)		
OAV9	344/50	311/39	3/8	1	(351.1)	(46.3)	(356.4)	(25.5)	(351.1)	(–3.7)	(20.2)	(28.2)
OAV10	352/53	322/37	2/8	2	(196.0)	(–55.3)	(204.6)	(–23.9)	(196.5)	(–4.0)		
OAV11	347/60	323/45	2/8	0	(–2.2)	(56.3)	(16.7)	(27.1)	(8.9)	(–2.9)		
OAV12	352/68	335/49	5/8	1	13.6	60.9	25.1	29.6	17.1	–5.6	294.0	5.4
OAV13	359/55	331/35	6/8	3	189.2	–53.7	200.0	–23.3	195.3	2.0	103.1	6.6
OAV14	356/20	258/22	5/8	0	357.3	56.0	14.1	27.9	1.1	35.4	146.7	6.3
OAV15	359/14	242/34	8/8	0	0.6	58.6	17.3	29.8	352.9	38.3	75.4	6.4
OAV16	007/70	353/44	6/7	3	200.3	–57.6	208.0	–25.6	204.9	11.7	52.5	9.3
Mean (in situ)			8/13	2	**6.6**	**57.5**					**240.1**	**3.6**
Mean (preplunge)			8/13	2			18.9	27.6			240.1	3.6
Mean (stratigraphic)			8/13	2					11.7	22.2	9.7	18.7
Magnetite Magnetization (Hanging wall)												
OAV6	056/50	079/20	9/9	0	333.8	40.6	351.0	21.9	358.7	19.8	108.1	5.0
OAV7	048/24	197/10	9/9	0	336.2	34.0	349.2	15.2	347.5	24.0	64.3	6.4
OAV8	090/30	157/26	7/8	0	329.8	23.0	338.5	8.6	338.8	34.6	46.3	9.0
OAV17*	060/33	133/11	8/10	0	337.0	34.7	350.1	15.5	352.3	24.7	198.3	5.0
OAV18*	322/110	331/100	10/10	5	134.5	–27.6	149.1	–20.0	140.8	78.9	28.1	9.3
OAV19	115/28	169/37	7/8	0	335.6	36.6	350.1	17.7	350.8	54.7	72.5	7.1
OAV20	107/36	150/39	5/8	1	323.8	31.2	338.4	18.4	344.6	56.7	87.2	8.2
Mean (in situ)			7/7	1	329.9	32.7					78.5	6.9
Mean (preplunge)			7/7	1			**343.8**	**16.9**			**78.5**	**6.9**
Mean (stratigraphic)			7/7	1					350.3	44.0	8.5	22.0

Notes: Abbreviations are the same as in Table 1. DD and D are the in situ dip direction and dip of bedding at the site. DD′ and D′ are the dip direction and dip of bedding corrected for the northeast structural plunge at Bare Mountain. Values in parentheses were not used in overall mean calculations. Bold values indicate the best estimate of the grand mean directions for the Antelope Valley Formation H and M component magnetizations at Bare Mountain.

*Collected as hand samples and cored in the lab.

tent with the seven site means. They were not considered further, however, because with only two sample directions for the silicic dikes, site-mean statistical parameters could not be determined. Normal, north and down directions were antipodal (within ~6°) to the reversed, south and up directions (Fig. 4). In addition, the two sites from diorite dikes (Ti9 and Ti10) yielded directions similar to those from the Miocene silicic porphyry dikes.

Antelope Valley Formation at Bare Mountain, Footwall Block

Natural remanent magnetization intensities for the Antelope Valley carbonates from the footwall rocks of the Meiklejohn Peak thrust ranged between 0.1 and 2.0 mA/m. Thermal demagnetization of samples revealed either a single or a dual component magnetization (Fig. 5, A and B). For samples with two components, the first component removed was a present-day field component (acquired by the samples in the recent geomagnetic field) with unblocking temperatures to about 350 °C (Fig. 5A). Because this component was likely acquired from recent weathering or is a recent viscous remanent magnetization, it was not deemed important to the discussion of the tectonic setting of Bare Mountain.

The characteristic remanent magnetization of the footwall rocks showed a moderate to steep north and down or antipodal south and moderate to steep up directions (in situ coordinates) with unblocking temperatures to 630 °C. High coercivities in excess of 200 mT were observed from AF (Fig. 5C) and combined AF and thermal demagnetizations. The high coercivity and unblocking temperatures above 585 °C suggested that hematite carries the characteristic remanent magnetization in the footwall samples; thus, in the following discussions this characteristic remanent magnetization is referred to as the H component.

The H component was isolated in samples from 8 of the 13 footwall sites (Table 2). In three of the remaining sites (OAV5, OAV10, and OAV11) remanence intensities were too weak to isolate the characteristic remanent magnetization in more than one or two samples. In the other two remaining sites (OAV3 and OAV9), directions were obtained from three samples, but the 95% confidence intervals about the sites means were excessively large ($\alpha_{95} > 25°$), and thus they were excluded from further consideration.

For all other samples from these sites, remanence intensities above 300 °C were near the background level of the cryogenic magnetometer (~5×10^{-7} A/m). Although results from the sites with weak remanence intensity revealed results consistent with the other sites, only those site mean directions defined by at least three sample directions were used to obtain an overall mean direction (Table 2).

Site mean directions from the eight footwall sites were well clustered about a north and moderate to steep down direction or an antipodal south and steep up direction (Fig. 6A). The characteristic remanent magnetization directions became north-northeast and moderately up after correction for the structural plunge of Bare Mountain and became progressively more dispersed as bedding was restored to paleohorizontal. The incremental fold test confirms that the H component magnetization was acquired after folding of these strata (Fig. 7A). This observation is based on the assumption that optimal clustering of the directions during unfolding, defined either by the minimum *f* value or peak κ value, indicates the position of the strata when the rocks acquired their magnetization (Graham, 1949).

Antelope Valley Formation at Bare Mountain, Hanging-Wall Block

Natural remanent magnetization intensities for the Antelope Valley carbonates from the hanging-wall rocks of the Meiklejohn Peak thrust ranged between 0.1 and 1.0 mA/m. Thermal and AF de-

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

magnetization typically yielded two components of magnetization. A present-day field component with north and steep down directions was unblocked between room temperature and 200 °C (Fig. 5D) or between the natural remanent magnetization and 15 mT (Fig. 5F). Following removal of the present-day field component, a characteristic remanent magnetization with northwest and moderate or antipodal southeast and moderate up directions (in situ coordinates) was isolated. Unblocking temperatures for this characteristic remanent magnetization were between 300 °C and 510 °C. Coercivities ranged between about 20 and 80 mT. This range of unblocking temperatures and coercivities, along with anhysteretic remanence and isothermal remanence magnetization acquisition curves, suggested that magnetite is the principal carrier of magnetization in these samples. Thus, we refer to this magnetization as the M component characteristic remanent magnetization in the following discussions.

Site mean directions for the M component magnetization were obtained from all seven hanging-wall sites (Table 2). Six of the site mean directions had northwest and moderate down directions. The seventh site (OAV18) had an equal number of southeast and up and antipodal northwest and down directions. The site mean directions were well clustered in in situ coordinates (Fig. 6B). Characteristic remanent magnetization directions moved to shallower inclinations upon correction for Bare Mountain plunge and dispersed after bedding was restored to paleohorizontal. An incremental fold test of the Meiklejohn Peak fold sites also showed that the M component magnetization was acquired after folding (Fig. 7B).

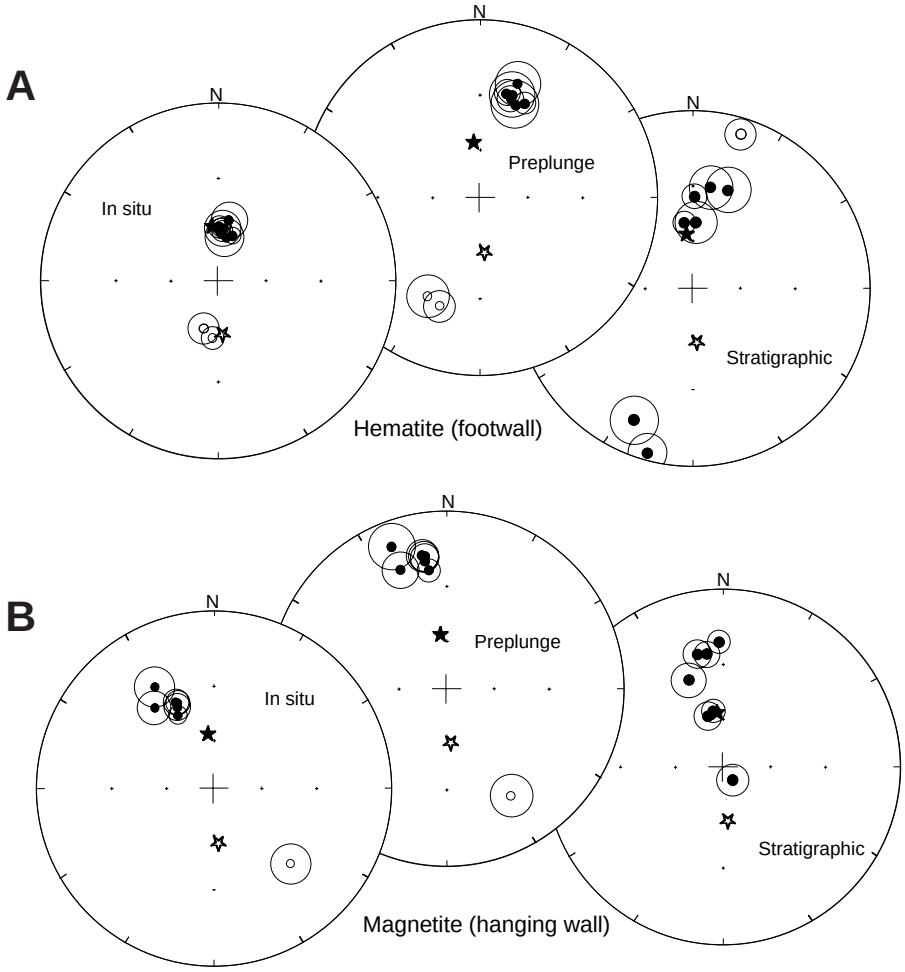


Figure 6. Equal-angle projection of site mean directions and associated 95% confidence regions for sites from (A) the footwall and (B) the hanging wall of the Meiklejohn Peak thrust in in situ, plunge-corrected, and tilt-corrected coordinates. Solid and open stars show the positions of the normal and reversed polarity reference directions, respectively, for Nevada based on the Miocene reference pole for North America (Hagstrum et al., 1987). Open and closed symbols are projections onto the upper and lower hemispheres, respectively.

Ordovician Pogonip Group Carbonate Rocks

The magnetizations of samples from sites distributed in the region around Bare Mountain (Fig. 1B, p. 1536) showed remanence characteristics similar to those of the samples from Bare Mountain. Samples from nine of these sites (Table 3), including all five from the Striped Hills, contained a high coercivity and high unblocking temperature characteristic remanent magnetization (Fig. 8, A, B, and C) similar to the H component found in the footwall rocks at Bare Mountain. Characteristic remanent magnetization directions from those sites had north and moderate to steep down directions or antipodal south and moderate to steep up directions (Fig. 9, A and B). Directions dispersed after unfolding, suggesting that the magnetizations postdate folding at these sites (Fig. 10). Unfortunately, for these five sites at the Striped Hills, bedding attitudes sites were not sufficiently different enough to perform a statistically meaningful fold test.

In contrast, samples from three sites show moderate unblocking temperatures (Fig. 8, D and E) and moderate coercivity spectra (Fig. 8E) similar to the M component in the hanging-wall rocks at Bare Mountain (Table 3). Directions from these sites have south-southeast declinations and inclinations that range between –22° and –63° (Fig. 9C). After tilt correction, however, the three site mean directions showed a common moderate shallow up inclination with east-southeast to south-southwest declinations. Inclination-only fold tests suggested that acquisition of the M component magnetization at three sites was prior to tilting of the blocks (Fig. 10).

Samples from the remaining two regional sites (OP8 and OP11) did not yield consistent site mean directions. Although individual samples appeared to have coherent directions, sample directions were dispersed along great circle trends and the resulting site-mean statistical parameters yielded unacceptably large α_{95} values (Table 3). We interpret these directions as composite magnetizations consisting of both normal and reversed polarity components with strongly overlapping coercivity and unblocking spectra. The resulting composite directions fall between a north and steep down component and an up and steep south component (OP11) or a southeast and moderate up component and a northwest and moderate down component (OP8). As with the

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

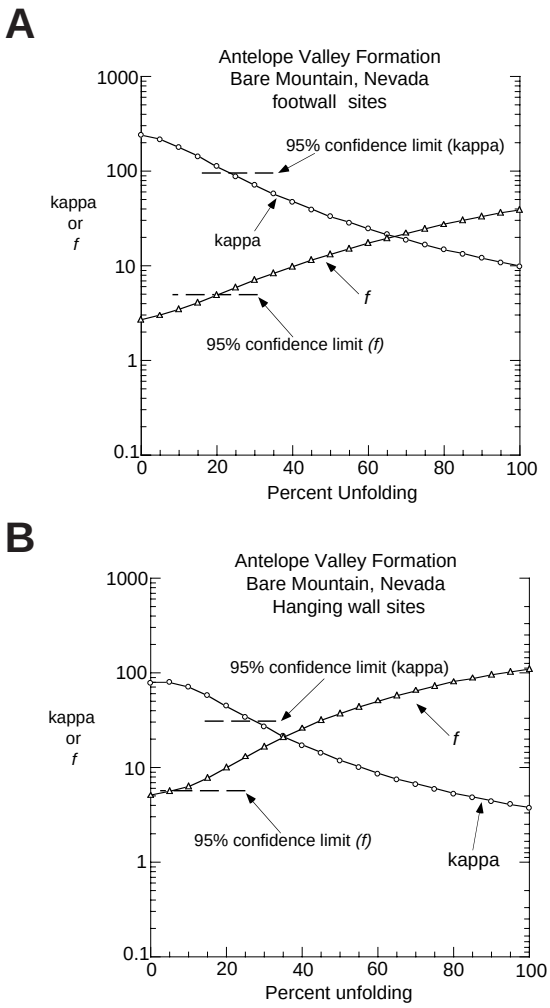


Figure 7. Incremental fold test diagrams showing variation of the McFadden and Jones (1981) fold test parameter (*f*) and the Fisher (1953) precision parameter (kappa) as a function of percent unfolding for the (A) footwall and (B) hanging-wall sites of the Meiklejohn Peak thrust. Higher values of kappa or lower values of *f* correspond to relatively better grouped directions. Dashed lines show the 95% significance levels for each test.

Miocene silicic porphyry dikes, remagnetization circle analyses were not performed because sufficient results could be obtained from those sites with stable, discrete-component sample behavior.

DISCUSSION

Origin of Characteristic Magnetizations in the Tertiary Dikes

There are two possible origins for the characteristic magnetization in the silicic porphyry dikes. The characteristic remanent magnetization is either a primary thermoremanent magnetization related to cooling of the dikes at about 14.5 Ma or a secondary chemical remanent magnetization carried by hydrothermal and epithermal alterations of the silicic dikes between 11.0 and 12.5 Ma. A third possibility, in which the characteristic remanent magnetization is a recent viscous remanent magnetization or a partial thermal-viscous remanent magnetization, acquired in the present dipole field (PDF), is unlikely. First, the magnetization has both polarity directions, dominated by reversed polarity directions—a situation uncommon for recently acquired viscous remanent magnetizations. Viscous remanent magnetizations typically have present-day field, normal polarity directions. Second, the characteristic remanent magnetization has unblocking temperatures to 580 °C, yet apatite fission track and isotopic dates show that the dikes remained cool, below about 150 °C, since about 12.5 Ma (Ferrill et al., 1996b; Weiss, 1996). According to the thermal activation relationships outlined in Pullaiah et al. (1975), the acquisition time required for a partial thermal-viscous remanent magnetization in magnetite at 150 °C is in excess of 10^8 yr.

Because hydrothermal and epithermal alterations occurred soon after intrusion of the dikes in Miocene time (Weiss, 1996; Noble et al., 1991), the question of a primary or secondary origin of the characteristic remanent magnetization cannot be easily resolved. The contact test did not yield coherent directions, which is unfortunate because that test may have constrained the timing of acquisition of dike magnetization. Zhang et al. (1993) reported that they obtained a positive baked-contact test and concluded that the dike remanence is primary, but those results have not been published. Ultimately, however, resolution of this ambiguity is not critical to the discussion of tectonic implications of the characteristic remanent magnetization directions because the difference in age between dike intrusion and hydrothermal or epithermal alterations is only a few million years and there has been little change in the reference polar wander path for Nevada during this interval. For this paper, we assumed that the characteristic remanent magnetization is a mid-Miocene magnetization as young as 11.0 Ma. Precise determination of the origin of the characteristic remanent magnetization awaits more rigorous petrographic and geochemical investigations of the dikes and the chilled margins.

The remanence in the two Oligocene(?) diorite dikes in the northwest corner of Bare Mountain appears to be either a secondary chemical remanent magnetization acquired during hydrothermal or epithermal alteration or a secondary thermoremanent magnetization or partial thermal-viscous remanent magnetization acquired as the northwestern corner of Bare Mountain was exhumed and uplifted(?) in middle Miocene time. Like the silicic porphyry dikes, the diorite dikes contain a dual polarity magnetizations with unblocking temperatures near 585 °C and moderate coercivity spectra with a median destructive field of 20 mT. The difference between the diorite and silicic porphyry dikes is that the diorite dikes remained above about 250 °C (Ferrill et al., 1996b) and perhaps as hot as 450 °C (Hoisch et al., 1997) until middle Miocene time, so it is unlikely they retained a primary thermoremanent magnetization. For example, the Pullaiah relationships indicate that rocks heated to 450 °C for 10^6 yr would have an equivalent unblocking temperature in the lab of nearly 530 °C. Therefore,

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

TABLE 3. PALEOMAGNETIC RESULTS OF THE POGONIP GROUP CARBONATES FROM THE STRIPED HILLS, YUCCA FLAT, MERCURY VALLEY, SPECTER RANGE, PAHRUMP VALLEY, AND BLACK HILLS, NEVADA

						In situ		Preplunge		Stratigraphic			
Site	DD/D	DD′/D′	n/N	R	D	Dec	Inc	Dec	Inc	Dec	Inc	κ	α_{95}
Striped Hills													
OPI	338/122	359/127	8/9	0	H	355.3	56.9	34.4	34.4	57.6	–62.2	72.2	7.9
OP2	341/123	003/126	8/8	0	H	338.0	52.2	19.2	51.2	35.4	–70.9	65.7	6.9
OP2	327/124	352/136	8/8	0	H	336.2	58.9	27.4	55.9	44.8	–65.9	51.0	8.1
OP4	352/116	007/115	8/8	0	H	337.1	64.1	36.4	58.0	31.4	–51.0	*105.0*	5.4
OP5	338/146	021/142	8/8	0	H	7.4	68.9	53.5	50.0	103.7	–69.6	25.8	11.1
Mean (in situ)			5/5	0		**345.5**	**60.7**					**79.3**	**8.6**
Mean (preplunge)			5/5	0				34.3	52.9			79.3	8.6
Mean (stratigraphic)			5/5	0						52.7	–65.4	35.7	13.0
Mercury Valley													
OP6	094/14		5/16	2	H	23.1	56.3			39.8	49.8	57.6	10.2
Yucca Flat													
OP7	101/79		7/8	0	H	9.1	47.7			58.0	9.4	32.0	10.8
OP8	015/42		7/8	1	L	(329.4)	(31.3)			(337.4)	(–0.8)	(7.5)	(24.6)
Pahrump Valley													
OP9	304/57		6/8	6	L	126.8	–62.7			125.3	–5.8	162.3	5.3
OPIO	001/21		8/8	7	L	145.8	–28.3			149.9	–10.7	35.4	9.4
Black Hills													
OPII	205/32		6/8	*5*	H	(199.4)	(–71.4)			(357.5)	(–70.4)	(2.5)	(60.0)
Specter Range													
OP12	175/60		5/8	3	H	184.2	–51.9			340.0	–67.3	39.1	10.9
OP13	245/09		6/8	6	L	192.4	–22.0			189.1	–17.2	30.6	12.3

Notes: Abbreviations are the same as in Tables 1 and 2. Bold values indicate the best estimate of the grand mean direction for the Pogonip Group carbonate rocks at the Striped Hills.

all the site mean directions from the dikes have been averaged to obtain a single grand mean direction (Table 1; Fig. 11A).

Origin of the H Component Magnetization

The H component in the footwall rocks (with respect to the Meiklejohn Peak thrust) at Bare Mountain and regional sites OP6, OP7, OP11, and OP12 appears to be a secondary chemical remanent magnetization acquired in middle Miocene time. The mean direction of the H component in in situ coordinates coincides with the Miocene reference direction and with the (Cenozoic) intermediate component magnetizations noted elsewhere in Nevada (e.g., Gillett and Van Alstine, 1982) (cf. Figs. 6A and 9, A and B, with the intermediate component shown in Fig. 11B). The possibility that the magnetization is younger, acquired in Pliocene or Quaternary time cannot, however, be ruled out. The interpretation that this direction is of Miocene age is simply based on the presence of both polarity magnetizations, evidence for significant hydrothermal or epithermal alterations during middle Miocene time (Weiss, 1996), and the lack of evidence for significant thermochemical alterations since that time.

Origin of the M Component Magnetization

The M component characteristic remanent magnetization in the hanging wall of the Meiklejohn Peak thrust and from regional sites OP8, OP9, OP10, and OP13 is interpreted as a chemical remanent magnetization related to regional, late Paleozoic chemical remagnetization, as first proposed by Gillett and Van Alstine (1982). Demagnetization characteristics of the M component characteristic remanent magnetization are similar to those observed in other remagnetized carbonates in Nevada (e.g., Gillett and Geissman, 1993) and to many chemically remagnetized carbonates worldwide (e.g., McCabe and Elmore, 1989; Van der Voo et al., 1997). All have peak unblocking temperatures in the range 450 to 500 °C and moderate coercivities with a median destructive field between 10 and 20 mT. In addition, saturation anhysteretic remanence–isothermal remanence magnetization ratios for samples with this characteristic remanent magnetization are between 0.07 and 0.15, slightly lower than values considered indicative of carbonate rocks with an authigenic magnetite remanence, but significantly larger than values considered representative of a primary magnetizations in carbonate rocks (Jackson et al., 1993).

Mean directions of the M component characteristic remanent magnetization are also consistent with a Permian-Triassic magnetization, once the directions are corrected for tilting related to Cenozoic Basin and Range extension. For example, correction for the northeast structural plunge of Bare Mountain yields moderate shallow inclinations with northwest or antipodal southeast declinations that fall on the reference Permian-Triassic path for Nevada (Fig. 11A). The resulting directions are also consistent with previously published results from remagnetized carbonates in nearby ranges in Utah and Nevada (Table 4, Fig. 11B). Tilt-corrected directions from OP9, OP10, and OP13 also yield shallow south-southwest and up directions that match the reference Permian-Triassic directions, except that they appear to also show variable degrees of clockwise rotation (Fig. 9C) similar to the rotated directions along the southern portions of the Nevada and Las Vegas Ranges (Gillett and Van Alstine, 1982; Nelson and Jones, 1987) (Fig. 11B). The main difference between these results and those previously reported is the presence of both normal and reversed polarity directions; this is in contrast to the pervasive reversed polarity direction found in Paleozoic carbonate rocks in the other ranges of Nevada and Utah. However, un-

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

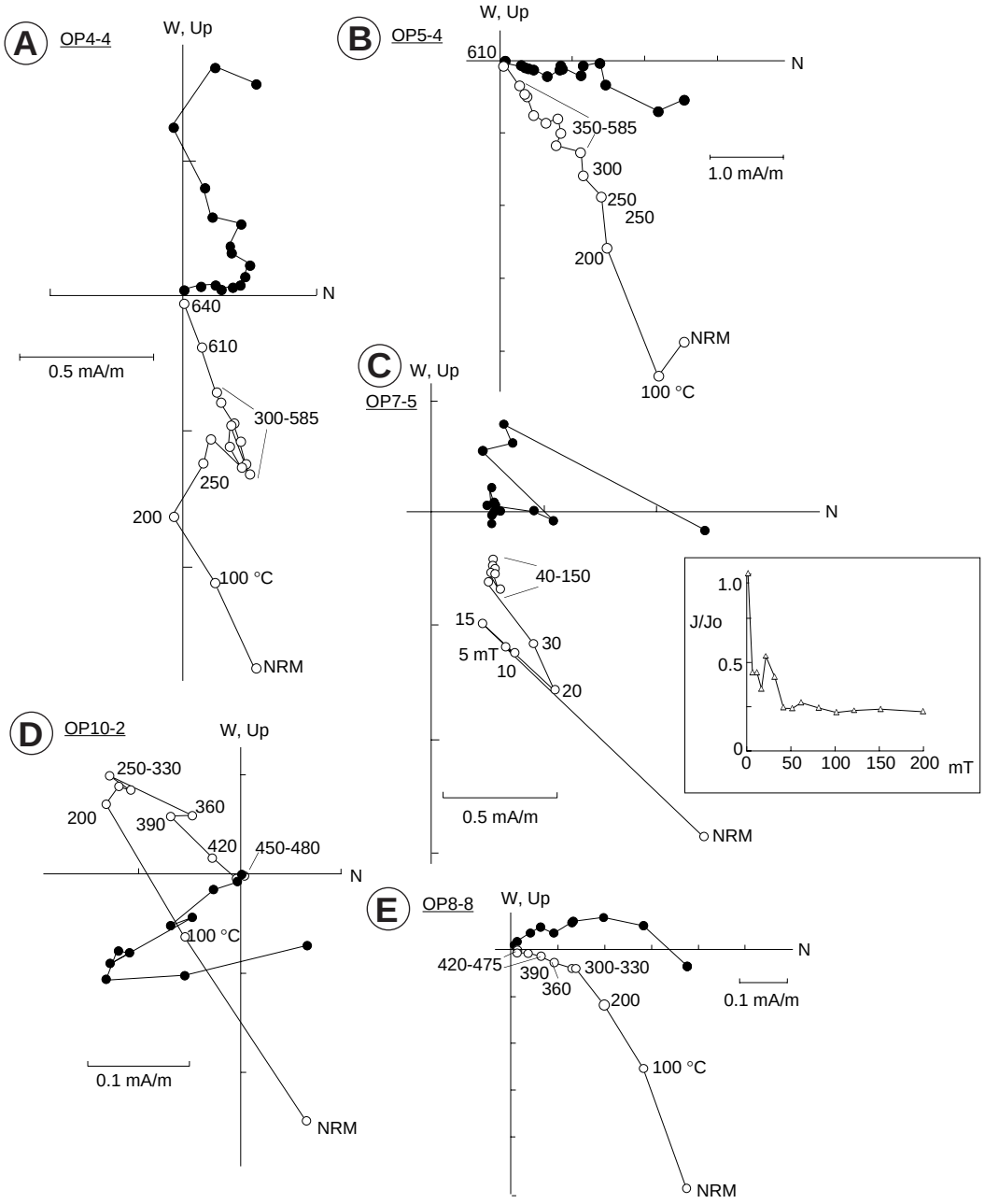


Figure 8. Sample vector end-point demagnetization diagrams showing thermal demagnetization of samples from Pogonip Group carbonates for regional sites around Bare Mountain. Open and closed symbols are projections onto the vertical and horizontal planes, respectively. In C, inset shows the variation of the normalized intensity (Jo—natural remanent magnetizations [NRM] intensity) as a function of applied alternating field.

like the mid-Permian B component in the central Appalachians (Stamatakos et al., 1996), the remagnetization in Basin and Range appears to postdate the end of the Kiaman superchron and instead corresponds to a period of active geomagnetic field reversals in latest Permian and Early Triassic time.

Relative Timing of Bare Mountain Tilting to Acquisition of Remanence

Tilting of Bare Mountain about a horizontal axis occurred sometime in middle to early Tertiary time, prior to eruption of the southwest Nevada volcanic field in middle Miocene time (Ferrill et al., 1996b). The inclinations of the characteristic remanent magnetization directions from the silicic and diorite dikes are consistent with this interpretation. For example, if the dike characteristic remanent magnetization directions are corrected for the structural plunge of Bare Mountain, then the resulting plunge-corrected di-

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

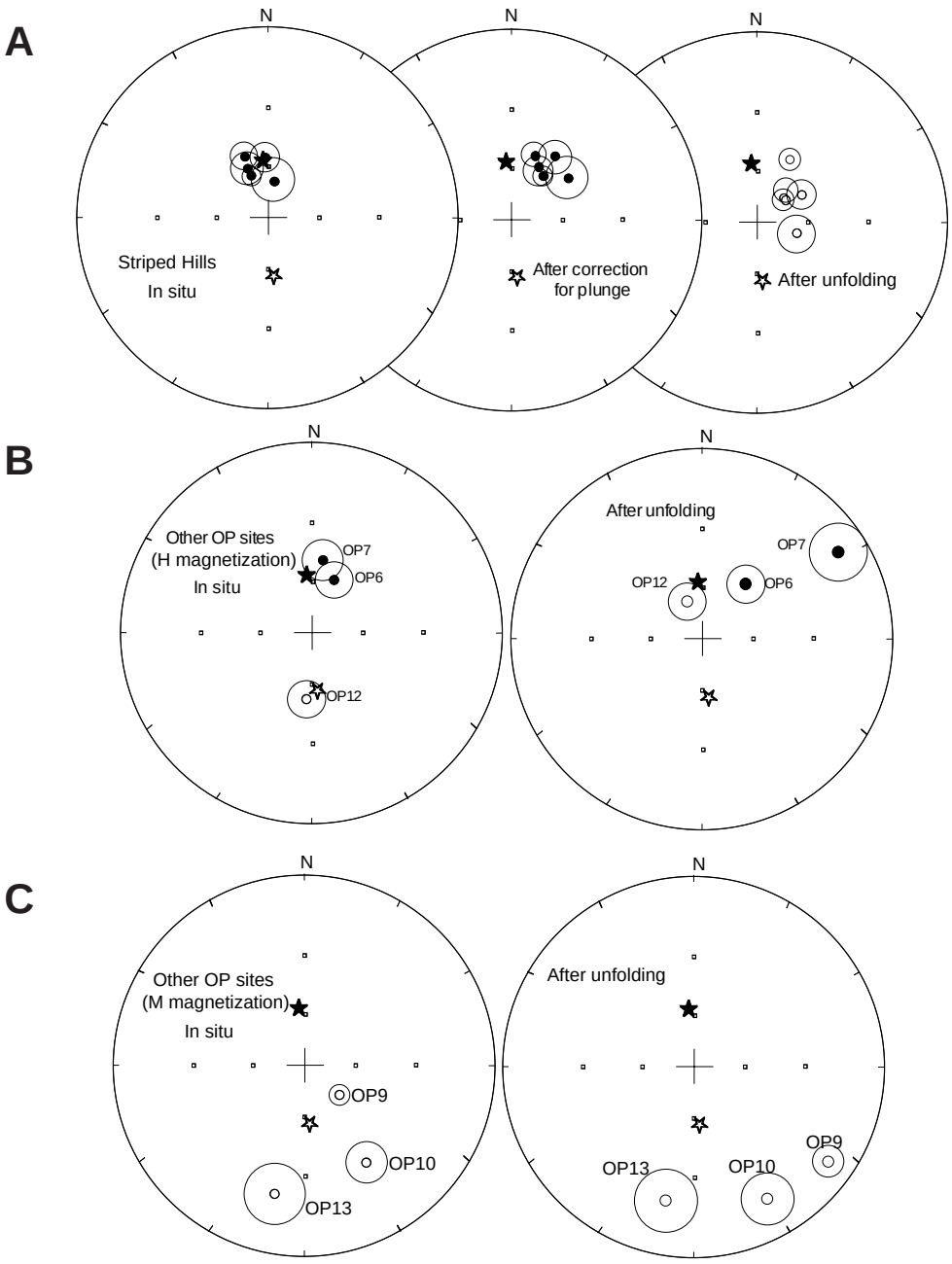


Figure 9. Equal-angle projections showing the site mean directions and associated 95% confidence regions for the Pogonip Group carbonates from (A) Striped Hills in in situ, plunge-corrected, and tilt-corrected coordinates, (B) hematite magnetization in in situ and tilt-corrected coordinates, and (C) magnetite magnetization in in situ and tilt-corrected coordinates. Solid and open stars show the position of the normal and reversed polarity reference directions, respectively, for Nevada, based on the Miocene reference pole for North America (Hagstrum et al., 1987). Open and closed symbols are projections onto upper and lower hemispheres, respectively.

rections have an inclination of 27.6° (Table 1; Fig. 4B) that is too shallow for any reasonable Tertiary magnetization at this site latitude in North America (expected inclination of 55° ± 3°). This result, coupled with the observation that all the silicic porphyry dikes are currently vertical, indicates that the dikes intruded Bare Mountain after the range was tilted about a horizontal axis.

Because of the good agreement between the H component characteristic remanent magnetization in the Antelope Valley carbonate rocks with the Miocene reference direction, the H component magnetization in the carbonate rocks appears to have been acquired after Bare Mountain and the Striped Hills were tilted. The mean directions for the footwall sites at Bare Mountain and the Striped Hills coincide in in situ coordinates, but diverge once they are corrected for the respective tilts of the two blocks (cf. Figs. 6A and 9A).

It is interesting that the observation that the H component magnetizations at the Striped Hills and Bare Mountain postdate tilting about horizontal axes of these ranges adds considerable complexity to the controversial proposed struc-

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

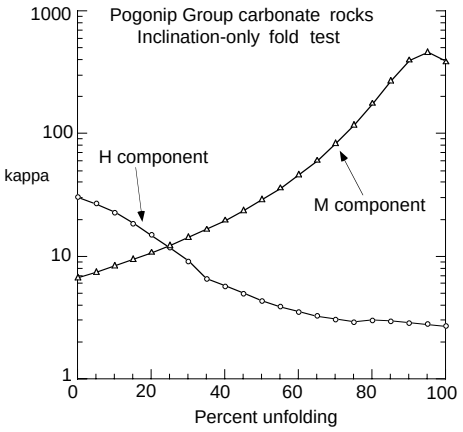


Figure 10. Inclination-only fold test showing variation of kappa as a function of percent unfolding for magnetite (M) and hematite (H) magnetizations in carbonate rocks from regional Pogonip Group sites.

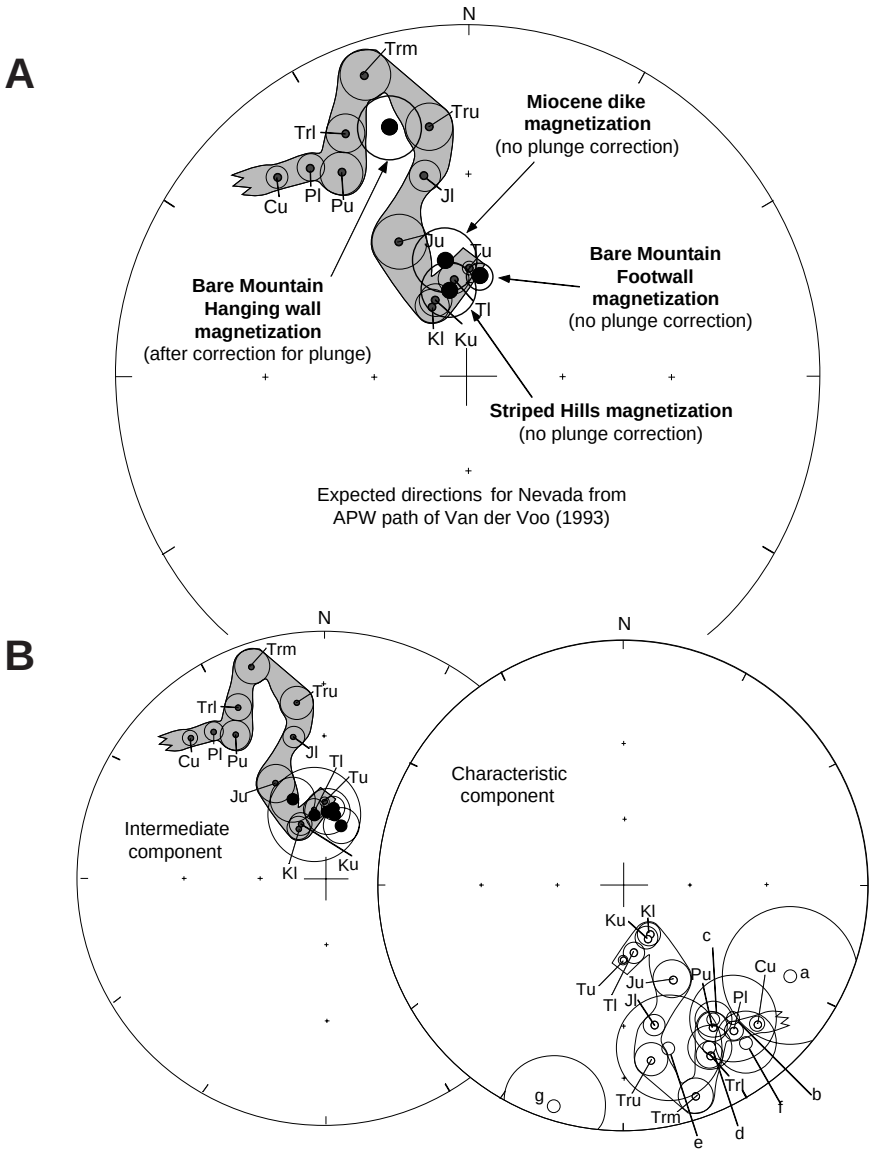


Figure 11. (A) Equal-angle projection of overall grand mean directions and associated 95% confidence intervals for Tertiary dikes, Striped Hills, and hanging wall and footwall of the Meiklejohn Peak thrust at Bare Mountain. Also shown are the reference directions from the Late Carboniferous to the Late Tertiary based on the apparent polar wander (APW) path for North America of Van der Voo (1993). (B) Equal-angle projection of characteristic and intermediate component magnetizations and associated 95% confidence regions from previously published paleomagnetic results of Paleozoic carbonate rocks in southern Nevada and western Utah for both intermediate and characteristic components. Mean characteristic directions are keyed by letter to Table 4. Open and closed symbols are projections onto upper and lower hemispheres, respectively.

tural link between Bare Mountain and the Striped Hills (Schweickert and Lahren, 1997). In that interpretation, the hanging-wall anticline of the Panama thrust at Bare Mountain is correlated across a major right-lateral strike-slip system (the Amargosa Desert fault) with a panel of vertical to overturned Paleozoic and Proterozoic rocks exposed in the Striped Hills (presumed to be the forelimb of an equivalent hanging-wall fold), under the assumption that both folds were once part of a single continuous structure in the Cordilleran Sevier fold-thrust belt. If correct, total right-lateral strike-slip offset of the fold would exceed 30 km. Yet, to achieve this realignment, the recognized eastward or northeastward tilts (plunges) of the Striped Hills and Bare Mountain must also be restored to horizontal. Otherwise, the folds in their present orientations cannot be restored along a strike-slip fault to form a single fold as proposed. This reconstruction would therefore require restoration of significant vertical offset (up to 7 km accounting for the 28° and 30° plunges and 15 km horizontal separation between Bare Mountain and the Striped Hills) in addition to the 30 km of strike-slip displacement. Because the characteristic remanent magnetization in the dikes at Bare Mountain and the H component in the carbonates at both Bare Mountain and the Striped Hills are no younger than about 11 Ma (and were acquired after tilting) and tilting of Bare Mountain about a horizontal axis is older than about 14 Ma (prior to intrusion of the silicic porphyry dikes), this vertical component of offset must be older than 14 Ma. Strike-slip motion on the proposed Amargosa Desert fault system is presumed to be younger than 12.5 Ma, occurring mostly after 11.5 Ma (Schweickert and Lahren, 1997), and thus it would postdate the vertical motion. Furthermore, apatite fission-track thermochronology indicates that the Tertiary thermal histories of the Striped Hills and southern Bare Mountain are incompatible (Stamatakos and Ferrill, 1996). Collectively, these complications, in addition to the ambiguous structural link between the Panama thrust fold and the overturned section at the Striped Hills, raise considerable doubt about the Schweickert and Lahren (1997) reconstruction.

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

TABLE 4. SUMMARY OF PREVIOUSLY PUBLISHED PALEOMAGNETIC DIRECTIONS FROM PALEOZOIC CARBONATES OF NEVADA AND WESTERN UTAH

			Intermediate					Characteristic				
Unit	Range	Ref	n	Dec	Inc	κ	α_{95}	n	Dec	Inc	κ	α_{95}
Carrara	Desert	a	3	349.4	60.4	39.4	19.9	3	119.4	–13.9	42.0	19.3
Bonanza King	Desert	a	6	1.2	59.4	47.8	9.8	7	141.4	–18.9	21.8	13.2
Nopah/Goodwin	Desert	a	2	7.4	60.7			3	146.7	–22.3	466.3	5.7
Pogonip Group	Egan	b						7	152.4	–15.7	89.2	6.4
CO Limestones	House	b	I*	*5.5*	58.0	21.1	*5.0*	4	164.6	–19.7	33.1	16.2
Bonanza King	Las Vegas	c	5†	337.2	51.3	69.1	9.3	*5*	142.5	–10.9	83.6	8.4
Bonanza King	Las Vegas	c	3§	15.2	64.6	252.0	7.8	3	197.0	–2.7	103.4	12.2

Notes: Abbreviations are the same as in Tables 1 and 2. n here is the number of sites averaged to obtain the formation mean direction.
References (Ref): a—Gillett and Van Alstine (1982); b—Gillett and Geissman (1993); c—Nelson and Jones (1987).
*Combined af and thermal directions with both polarities listed.
†Combined northern sites (NM, CNM, UP, LP, and YF).
§Combined southern sites (CK, LO, and IR).

The observation that the M component characteristic remanent magnetization fails the fold test at Bare Mountain also shows that folding of the Meiklejohn Peak thrust anticline occurred prior to acquisition of this magnetization. The shallow inclination of the M component magnetization and its similarity to other late Paleozoic secondary magnetizations in the Basin and Range indicate acquisition in Late Permian or Early Triassic time. This leads us to conclude that folding of the Meiklejohn Peak anticline occurred in late Paleozoic or early Mesozoic time, prior to Jurassic or Cretaceous Sevier deformation. This conclusion is further supported by zircon fission-track data, which yield Pennsylvanian-Permian cooling ages for the hanging-wall rocks of the Meiklejohn Peak thrust (Ferrill et al., 1996b). Thus, in spite of differences in the prediction of large-magnitude vertical-axis rotations (discussed below), the paleomagnetic results from this study support Snow's (1992) hypothesis that this portion of the Basin and Range, contains pre-Sevier (Permian or older) contractional structures.

Vertical-Axis Rotations

Comparison of the chemical remanent magnetization directions in the dikes to the reference Miocene direction for Nevada shows no evidence for large-scale rotations of Bare Mountain. The data suggest a small, but statistically insignificant, counterclockwise rotation of –11° ± 13°. The expected reference direction for this part of Nevada is D = 357°, I = 55°, DI = 3.0°, DD = 3.7°, based on the Miocene reference pole of Hagstrum et al. (1987). The amount of counterclockwise rotation would be slightly greater (~2°) if it also included a small correction for the tilt of Bare Mountain related to post-Miocene motion along the Bare Mountain fault.

The small counterclockwise rotation is surprising given the prediction that Bare Mountain rotated clockwise, if at all. The dual polarity magnetization suggests that the magnetization was acquired over one or more reversals of the Earth's magnetic field and, thus, the counterclockwise rotation does not appear to be an artifact of inadequately averaged secular variation of the Earth's geomagnetic field. This is especially true if the dike's characteristic remanent magnetization is a chemical remanent magnetization or partial thermal-viscous remanent magnetization, which requires an acquisition time significantly longer than 5–10 k.y. Given the noted problems in isolating a single component direction from the two antipodal components in these rocks, however, it is possible this apparent counterclockwise rotation reflects a slight bias in the characteristic remanent magnetization directions due to unremoved components of opposite polarity magnetizations. In addition, the moderate to steep inclinations (50°–60°) may also obscure accurate quantification of the small-angle rotations in these rocks.

Comparison of the H component of magnetization with the Miocene reference direction also reveals negligible rotations. The grand mean direction from Bare Mountain indicates a small clockwise rotation of 10° ± 6° compared to a statistically insignificant counterclockwise rotation of –12° ± 14° for the grand mean direction at the Striped Hills (Table 3). Although it is possible that these values represent actual vertical-axis rotations of these blocks, the moderate to steep inclinations and small angles make us uneasy about assigning undue significance to these values.

Comparison of the M component magnetization at Bare Mountain also indicates only a minor vertical-axis rotation of Bare Mountain, if any. The plunge-corrected direction for the M component falls within the Triassic portion of the reference path for southern Nevada based on the North American apparent polar wander path of Van der Voo (1993). Because the regional data for Nevada and Utah suggest that remagnetization is most likely Late Permian or Early Triassic in age (e.g., Gillett and Van Alstine, 1982), the Late Permian reference pole (Van der Voo, 1993) was used to derive the reference direction of D = 328°, I = 23°, DI = 5.4°, DD = 4.1°. Comparison of the mean M component characteristic remanent magnetization at Bare Mountain with that direction yields a 16° ± 7° clockwise rotation. Comparison of the M component of magnetization to younger Triassic reference directions yields statistically concordant declinations (Fig. 11A).

Although paleomagnetic results from Bare Mountain indicate small clockwise and counterclockwise rotations, the data clearly do not support the ~90° clockwise rotation of Bare Mountain suggested by Snow (1992) and Snow and Prave (1994). In addition, the paleomagnetic results at Bare Mountain contrast with the post mid-Miocene 30°–50° clockwise rotations derived from paleomagnetic results of the Miocene tuffs north, south, and east of the range. Thus, even though Bare Mountain was tilted about a horizontal axis during exhumation and uplift, it appears to have remained relatively fixed about its vertical axis since at least the late Paleozoic or early Mesozoic time. We suggest that one way to accommodate the relatively large rotations observed in the tuffs just south and east of Bare Mountain is by steep-axis rotations related to extension along normal faults with the large displacement gradients on the intrabasinal normal faults in Crater Flat (e.g., Rosenbaum et al, 1991; Stamatakos and Ferrill, 1998, in press).

Only sites OP9 and OP10 appear to show significant vertical-axis rotations in our analysis. Because of the limited number of sites, it is not clear how the rotations here relate to overall Walker Lane transtension and Basin and Range extension. The presence of what appears to be Permian-Triassic magnetization preserved in these rocks, however, makes them ideal candidates for future paleomagnetic investigations in and around the Pahrump Valley.

CONCLUSIONS

On the basis of the paleomagnetic results presented in this paper, we reach the following conclusions.

1. The Ordovician carbonates of the Antelope

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

Valley Formation at Bare Mountain contain two distinct secondary but ancient magnetizations. Samples from the hanging wall of the Meiklejohn Peak thrust have a characteristic remanent magnetization carried by authigenic magnetite related to regional remagnetization of southern Nevada in Late Permian or Early Triassic time. Samples from the footwall of the Meiklejohn Peak thrust have a Miocene-age chemical remanent magnetization carried by hematite related to hydrothermal or epithermal alterations associated with magnetizations in the southwestern Nevada volcanic field. The acquisition of the Permian-Triassic magnetization postdates folding of the Meiklejohn Peak anticline, suggesting that the Meiklejohn Peak thrust relates to Permian or earlier compression.

2. The Tertiary mafic and silicic porphyry dikes at Bare Mountain contain a dual-polarity Miocene magnetization that is either a primary thermoremanent magnetization or a secondary chemical remanent magnetization related to hydrothermal or epithermal alterations associated with development of the southwestern Nevada volcanic field.

3. On the basis of the characteristic remanent magnetization directions in all three magnetizations, Bare Mountain has not significantly rotated about a vertical axis since late Paleozoic or early Mesozoic time.

4. The age of Basin and Range tilting about horizontal axes at Bare Mountain and the Striped Hills appears to predate acquisition of the secondary chemical remanent magnetization carried by hematite and the primary or secondary magnetization in the diorite and silicic dikes at Bare Mountain. This age constraint raises doubts about the proposed reconstruction of a once-continuous and juxtaposed Bare Mountain and Striped Hills fold across the Amargosa Desert fault system.

ACKNOWLEDGMENTS

This paper results from work performed by the Center for Nuclear Waste Regulatory Analyses (CNWRA) for the U.S. Nuclear Regulatory Commission (NRC) under contract number NRC-02-93-005. The activities reported here were initiated on behalf of the NRC Office of Research and continued under the Office of Nuclear Material Safety and Safeguards. This paper is an independent product of CNWRA and does not necessarily reflect the views or regulatory positions of the NRC. We thank Barbara Long for editorial reviews, Annette Mandujano for assistance in manuscript preparation, and Larry McKague, Mike Conway, Wesley Patrick, John Trapp, Philip Justus, and Bakr Ibrahim for technical comments and reviews. GSA *Bulletin* reviews by L. Brown, R. D. Elmore, and especially S. S. Harlan greatly improved this paper. We also thank Sid Jones, Bret Rahe, and Don Cederquist for their help in sample collection and laboratory analyses.

REFERENCES CITED

Albers, A. N., and Stewart, J. H., 1972, Geologic and mineral deposits of Esmeralda County, Nevada: Nevada Bureau of Mines and Geology Bulletin 78, 80 p.

Carr, W. J., 1988, Volcano-tectonic setting of Yucca Mountain and Crater Flat, southwestern Nevada, *in* Carr, M., and Yount, J., eds., Geologic and hydrologic investigations of a potential nuclear waste disposal site at Yucca Mountain, southern Nevada: U.S. Geological Survey Bulletin 1790, p. 35–49.

Caskey, S. J., and Schweickert, R. A., 1992, Mesozoic deformation in the Nevada test site and vicinity: Implications for the structural framework of the Cordilleran fold and thrust belt and Tertiary extension north of Las Vegas Valley: Tectonics, v. 11, p. 1314–1331.

Cornwall, H. R., 1972, Geology and mineral deposits of Nye County, Nevada: Nevada Bureau of Mines and Geology Bulletin 77, p. 49.

Cornwall, H. R., and Kleinhample, F. J., 1961, Geology of the Bare Mountain quadrangle: U.S. Geological Survey Quadrangle Map GQ-157, scale 1:62 500, 1 sheet.

Demarest, H. H., 1983, Error analysis for the determination of tectonic rotations from paleomagnetic data: Journal of Geophysical Research, v. 88, p. 4321–4328.

Ferrill, D. A., Stamatakos, J. A., Jones, S. M., Rahe, B., McKague, H. L., Martin, R. H., and Morris, A. P., 1996a, Quaternary slip history of the Bare Mountain Fault (Nevada) from the morphology and distribution of alluvial fan deposits: Geology, v. 24, p. 559–562.

Ferrill, D. A., Stirewalt, G. L., Henderson, D. B., Stamatakos, J. A., Spivey, K. H., and Wernicke, B. P., 1996b, Faulting in the Yucca Mountain region, critical review and analyses of tectonic data from the central Basin and Range: Washington, D.C., Nuclear Regulatory Commission, Report NUREG/CR-6401.

Ferrill, D. A., Stamatakos, J. A., and McKague, H. L., 1997, Quaternary slip history of the Bare Mountain Fault (Nevada) from the morphology and distribution of alluvial fan deposits: Reply: Geology, v. 25, p. 190.

Ferrill, D. A., Morris, A. P., Jones, S. M., and Stamatakos, J. A., 1998, Extensional layer-parallel shear and normal faulting: Journal of Structural Geology, v. 20, p. 355–362.

Fisher, R. A., 1953, Dispersion on a sphere: Proceedings of the Royal Society of London, ser. A, v. 217, p. 295–305.

Fridrich, C., 1998, Tectonic evolution of the Crater Flat basin, Yucca Mountain, Nevada: U.S. Geological Survey Open-File Report 98-33, 43 p.

Gillett, S. L., and Geissman, J. W., 1993, Regional late Paleozoic remagnetization of shallow-water carbonate rocks in the Great Basin: The usefulness of "micro"-fold tests from late compaction fabrics, *in* Application of paleomagnetism to sedimentary geology: Society for Sedimentary Geology (SEPM) Special Publication 49, p. 129–147.

Gillett, S. L., and Van Alstine, D. R., 1982, Remagnetization and tectonic rotation of upper Precambrian and lower Paleozoic strata from the Desert Range, southern Nevada: Journal of Geophysical Research, v. 87, p. 10929–10953.

Graham, J. W., 1949, The stability and significance of magnetism in sedimentary rocks: Journal of Geophysical Research, v. 54, p. 131–167.

Grow, J. A., Barker, C. E., and Harris, A. G., 1994, Oil and gas exploration near Yucca Mountain, southern Nevada, *in* Proceedings, Fifth Annual International Conference on High-Level Radioactive Waste Management: La Grange Park, Illinois, American Nuclear Society International, p. 1298–1315.

Hagstrum, J. T., Sawlan, M. G., Hausback, B. P., Smith, J. G., and Grommé, C. S., 1987, Miocene paleomagnetism and tectonic setting of the Baja California Peninsula, Mexico: Journal of Geophysical Research, v. 92, p. 2627–2639.

Hoisch, T. D., Heizler, M. T., and Zartman, R. E., 1997, Timing of detachment faulting in the Bullfrog Hills and Bare Mountain area, southwest Nevada, inferences from $^{40}Ar/^{39}Ar$, K-Ar, U-Pb, and fission track thermochronology: Journal of Geophysical Research, v. 102, p. 2815–2834.

Holm, D. K., Geissman, J. W., and Wernicke, B., 1993, Tilt and rotation of the footwall of a major normal fault system: Paleomagnetism of the Black Mountains, Death Valley extended terrane, California: Geological Society of America Bulletin, v. 105, p. 1373–1387.

Hudson, M. R., 1992, Paleomagnetic data bearing on the origin of arcuate structures in the French Peak–Massachusetts Mountains area of southern Nevada: Geological Society of America Bulletin, v. 104, p. 581–594.

Hudson, M. R., Sawyer, D. A., and Warren, R. G., 1994, Paleomagnetism and rotation constraints for the Miocene southwestern Nevada volcanic field: Tectonics, v. 13, p. 258–277.

Hudson, M. R., Minor, S. A., and Fridrich, C. J., 1996, The distribution, timing, and character of steep-axis rotations in a broad zone of dextral shear in southwestern Nevada: Geological Society of America Abstracts with Programs, v. 28, no. 7, p. A451.

Jackson, M., Rochette, P., Fillion, G., Banerjee, S., and Marvin, J., 1993, Rock magnetism of remagnetized Paleozoic carbonates: Low temperature behavior and susceptibility characteristics: Journal of Geophysical Research, v. 98, p. 6217–6225.

Kirschvink, J. L., 1980, The least-squares line and plane and the analysis of paleomagnetic data: Geophysical Journal of the Royal Astronomical Society, v. 62, p. 699–718.

Longwell, C. R., Pampeyan, E. H., Bower, B., and Roberts, R. J., 1965, Geology and mineral deposits of Clark County: Nevada, Nevada Bureau of Mines and Geology Bulletin 62, 218 p.

Maldonado, F., 1990, Structural geology of the upper plate of the Bullfrog Hills detachment fault system, southern Nevada: Geological Society of America Bulletin, v. 102, p. 992–1006.

Marvin, R. F., Mehnert, H. H., and Naeser, C. W., 1989, U.S. Geological Survey radiometric ages compilation "C," Part 3: California and Nevada: Isochron/West, no. 52, p. 3–11.

McCabe, C., and Elmore, R. D., 1989, The occurrence and origin of late Paleozoic remagnetization in the sedimentary rocks of North America: Reviews of Geophysics, v. 27, p. 471–494.

McElhinny, M., 1964, Statistical significance of the fold test in paleomagnetism: Geophysical Journal, v. 8, p. 338–340.

McFadden, P. L., and Jones, D. L., 1981, The fold test in paleomagnetism: Geophysical Journal of the Royal Astronomical Society, v. 67, p. 53–58.

McFadden, P. L., and Reid, A. B., 1982, Analysis of paleomagnetic inclination data: Geophysical Journal of the Royal Astronomical Society, v. 69, p. 307–319.

Monsen, S. A., Carr, M. D., Regeis, M. C., and Orkild, P. A., 1992, Geologic map of Bare Mountain, Nye County, Nevada: U.S. Geological Survey Miscellaneous Investigations Series Map I-2201, scale 1:24 000, 1 sheet, 6 p.

Nelson, M. R., and Jones, C. H., 1987, Paleomagnetism and crustal rotations along a shear zone, Las Vegas Range, southern Nevada: Tectonics, v. 6, p. 13–33.

Noble, D. C., Weiss, S. I., and Green, S. M., 1989, High-salinity fluid inclusions suggest that Miocene gold deposits of the Bare Mountain district, Nevada, are related to a large buried rare-metal rich magmatic system: Geological Society of America Abstracts with Programs, v. 21, no. 6, p. A123.

Noble, D. C., Weiss, S. I., and McKee, E. H., 1991, Magmatic and hydrothermal activity, caldera geology, and regional extension in the western part of the southwestern Nevada volcanic field, *in* Raines, G. L., Lisle, R. E., Shafer, R. W., and Wilkerson, W. W., eds., Geology and ore deposits of the Great Basin: Reno, Geological Society of Nevada, p. 913–934.

Oldow, J. S., Kohler, G., and Donelick, R. A., 1994, Late Cenozoic extensional transfer in the Walker Lane strike-slip belt, Nevada: Geology, v. 22, p. 637–640.

Parés, J. M., Van der Voo, R., Stamatakos, J., and Pérez-Estaún, A., 1994, Remagnetization and post-folding oroclinal rotations in the Cantabrian/Asturian Arc, northern Spain: Tectonics, v. 13, p. 1461–1471.

Pullaiah, G. E., Irving, E., Buchan, K. L., and Dunlop, D. J., 1975, Magnetization changes caused by burial and uplift: Earth and Planetary Science Letters, v. 28, p. 133–143.

Rosenbaum, J. G., Hudson, M. R., and Scott, R. B., 1991, Paleomagnetic constraints on the geometry and timing of

Downloaded from gsabulletin.gsapubs.org on October 28, 2010

deformation at Yucca Mountain, Nevada: Journal of Geophysical Research, v. 96, p. 1963–1979.

Sawyer, D., Fleck, R. J., Lanphere, M. A., Warren, R. G., Broxton, D. E., and Hudson, M., 1994, Episodic caldera volcanism in the Miocene southwestern Nevada volcanic field: Revised stratigraphic framework, $^{40}Ar/^{39}Ar$ geochronology and implications for magmatism and extension: Geological Society of America Bulletin, v. 106, p. 1304–1318.

Schweickert, R. A., and Lahren, M. M., 1997, Strike-slip fault system in Amargosa Valley and Yucca Mountain, Nevada: Tectonophysics, v. 272, p. 25–41.

Snow, J. K., 1992, Large-magnitude Permian shortening and continental-margin tectonics in the southern Cordillera: Geological Society of America Bulletin, v. 104, p. 80–105.

Snow, J. K., 1994, Mass balance of Basin and Range extension as a tool for geothermal exploration: Geothermal Resources Council Transactions, v. 18, p. 1–8.

Snow, J. K., and Prave, A. R., 1994, Covariance of structural and stratigraphic trends: Evidence for anticlockwise rotation within the Walker Lane Belt Death Valley region, California and Nevada: Tectonics, v. 13, p. 712–724.

Snow, J. K., Geissman, J. W., and Wernicke, B., 1993, Paleomagnetic data from Paleozoic sedimentary rocks and Permo-Triassic intrusions, Death Valley area, southeast CA: Implications for Mesozoic and Cenozoic deformation [abs.]: Eos (Transactions, American Geophysical Union), v. 74, p. 206.

Sonder, L. J., Jones, C. H., Salyards, S. L., and Murphy, K. M., 1994, Vertical axis rotations in the Las Vegas shear zone, southern Nevada: Paleomagnetic constraints on kinematics and dynamics of block rotations, Tectonics, v. 13, p. 769–788.

Stamatakos, J. A., and Ferrill, D. A., 1996, Kinematic constraints of central Basin and Range tectonism from paleomagnetic and fission track studies at Bare Mountain, Nevada: Geological Society of America Abstracts with Programs, v. 28, no. 7, p. A125.

Stamatakos, J. A., and Ferrill, D. A., 1998, Strike-Slip fault system in Amargosa Valley and Yucca Mountain, Nevada: Comment: Tectonophysics (in press).

Stamatakos, J., Hirt, A. M., and Lowrie, W., 1996, The age and timing of folding in the central Appalachians from paleomagnetic results: Geological Society of America Bulletin, v. 108, p. 815–829.

Stamatakos, J. A., Connor, C. B., and Martin, R. H., 1997, Quaternary basin evolution and basaltic volcanism of Crater Flat, Nevada, from detailed ground magnetic surveys of the Little Cones: Journal of Geology, v. 105, p. 319–330.

Stewart, J. H., 1988, Tectonics of the Walker Lane belt, western Great Basin: Mesozoic and Cenozoic deformation in a zone of shear, *in* Ernst, W. G., ed., Metamorphism and crustal evolution of the western United States: Englewood Cliffs, New Jersey, Prentice-Hall, p. 683–713.

Trexler, J. H., Jr., Cole, J. C., and Cashman, P. H., 1996, Middle Devonian–Mississippian stratigraphy on and near the Nevada test site: Implications for hydrocarbon potential: American Association of Petroleum Geologists Bulletin, v. 80, p. 1736–1762.

Tschanz, C. M., and Pampeyan, E. H., 1970, Geology and mineral deposits of Lincoln County, Nevada: Nevada Bureau of Mines and Geology Bulletin 73, 187 p.

Van der Voo, R., 1993, Paleomagnetism of the Atlantic, Tethys, and Iapetus Oceans: Cambridge, United Kingdom, Cambridge University Press, 411 p.

Van der Voo, R., Stamatakos, J., and Parés, J. M., 1997, Kinematic constraints on thrust-belt curvature from syndeformational magnetizations in the Lagos Del Valle syncline in the Cantabrian Arc, Spain: Journal of Geophysical Research, v. 102, p. 10105–10119.

Weiss, S. I., 1996, Hydrothermal activity, epithermal mineralization and regional extension in the southwestern Nevada volcanic field [Ph.D. dissert.]: Reno, University of Nevada, 212 p.

Wernicke, B. P., 1992, Cenozoic extensional tectonics of the western Cordillera, in Burchfiel, B. C., Lipman, P. W., and Zoback, M. L., eds., The Cordilleran orogen: Conterminous U.S.: Boulder, Colorado, Geological Society of America, Geology of North America, v. G-3, p. 553–581.

Zhang, Y., Gillett, S. L., Karlin, R. E., and Schweickert, R. A., 1993, Paleomagnetism of Miocene dikes in Bare Mountain, southwestern Nevada: Implications for regional tectonics: Geological Society of America Abstracts with Programs, v. 25, no. 5, p. A169.

Zijderveld, J. D. A., 1967, AC demagnetization of rocks: Analysis of results, *in* Runcorn, S. K., Creer, K. M., and Collinson, D. W., eds., Methods in paleomagnetism: Amsterdam, Elsevier Science, p. 254–286.

MANUSCRIPT RECEIVED BY THE SOCIETY JULY 17, 1997
REVISED MANUSCRIPT RECEIVED DECEMBER 22, 1997
MANUSCRIPT ACCEPTED FEBRUARY 8, 1998

Printed in U.S.A.